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- (54) **AURISTATIN DRUG LINKER CONJUGATES**
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This patent is subject to a terminal disclaimer.

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Related U.S. Application Data

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(51) **Int. Cl.**

A61K 39/44 (2006.01)
A61K 38/08 (2006.01)
A61K 47/48 (2006.01)
C07K 5/02 (2006.01)
A61K 45/06 (2006.01)
G01N 33/50 (2006.01)
A61K 38/00 (2006.01)

(52) **U.S. Cl.**

CPC **A61K 47/48723** (2013.01); **A61K 38/08** (2013.01); **A61K 45/06** (2013.01); **A61K 47/4863** (2013.01); **A61K 47/48338** (2013.01); **A61K 47/48415** (2013.01); **A61K 47/48561** (2013.01); **A61K 47/48607** (2013.01); **A61K 47/48623** (2013.01); **C07K 5/0205** (2013.01); **G01N 33/5008** (2013.01); **A61K 38/00** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

Drug Linker compounds and Drug Linker Ligand conjugates are provided that have auristatins linked via the C-terminus. The conjugates show efficacy without the need for a self-immolative group to release the drug.

43 Claims, 9 Drawing Sheets

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Figure 1

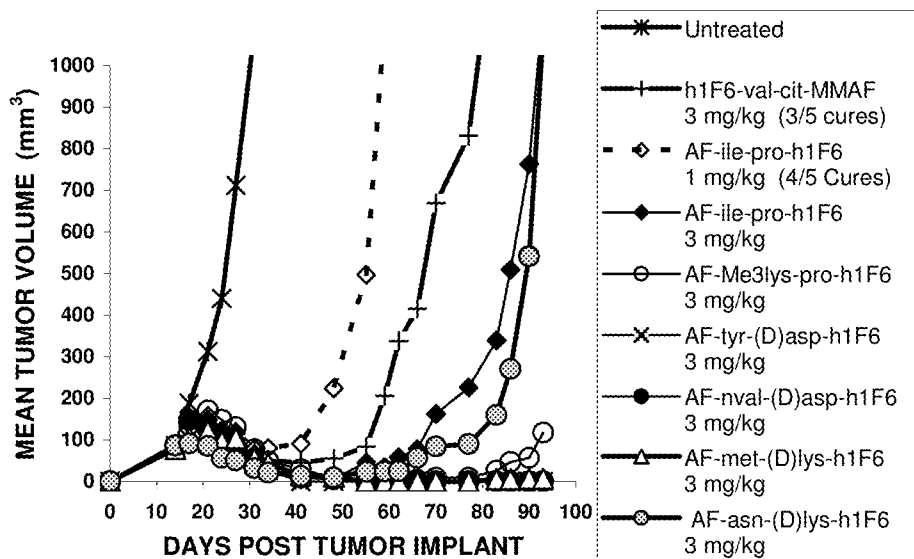


Figure 2

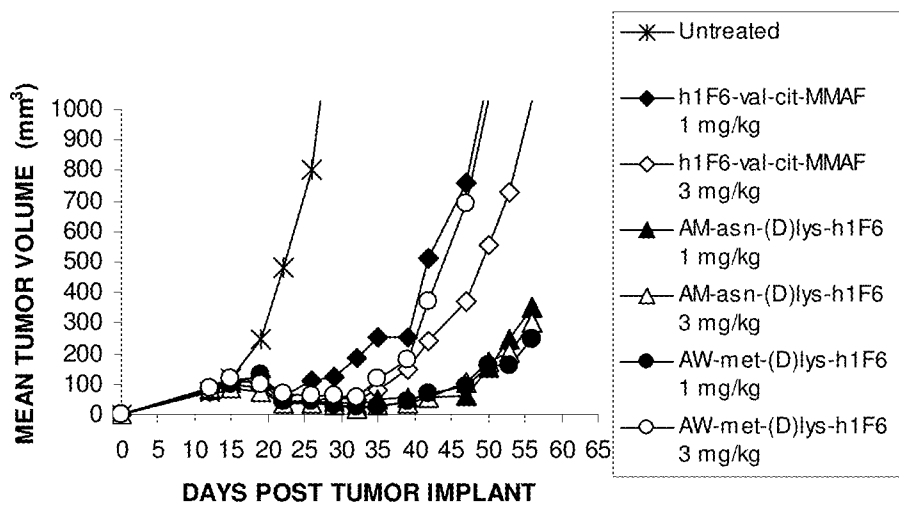


Figure 3

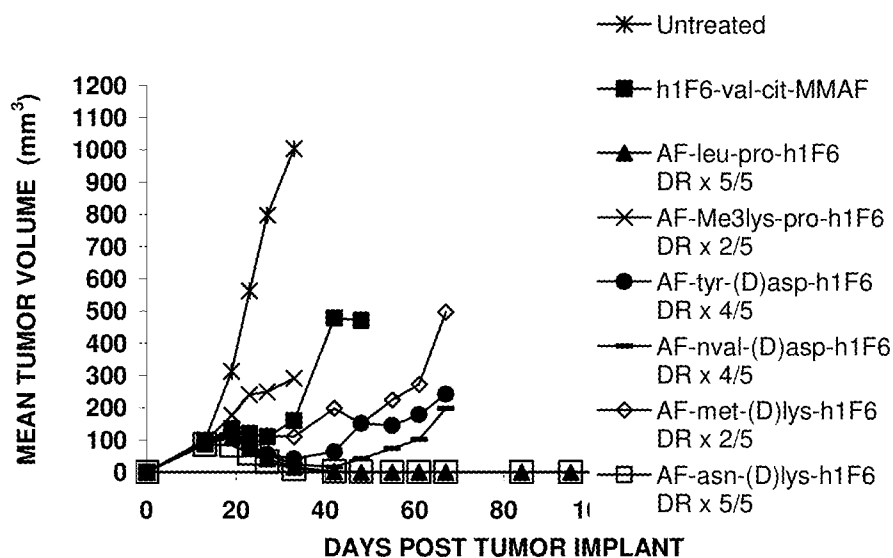


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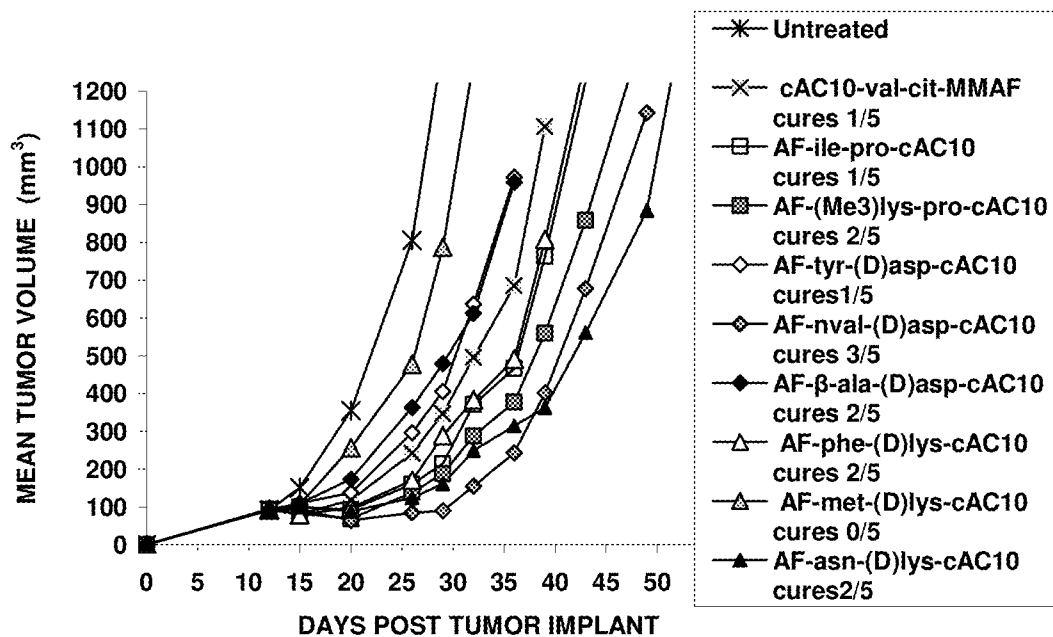


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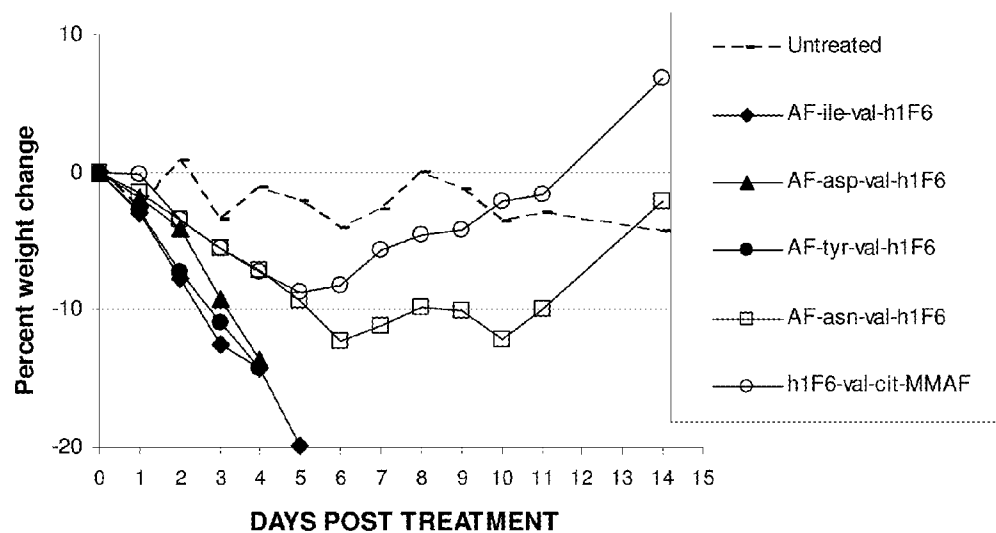


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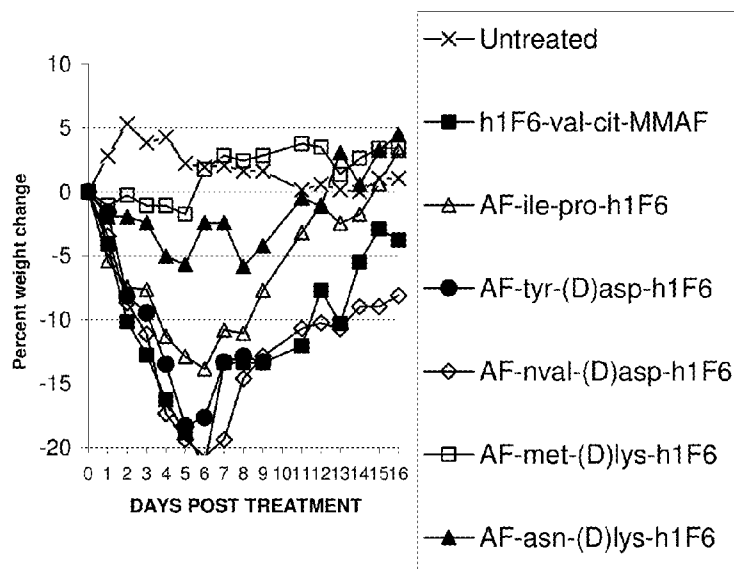


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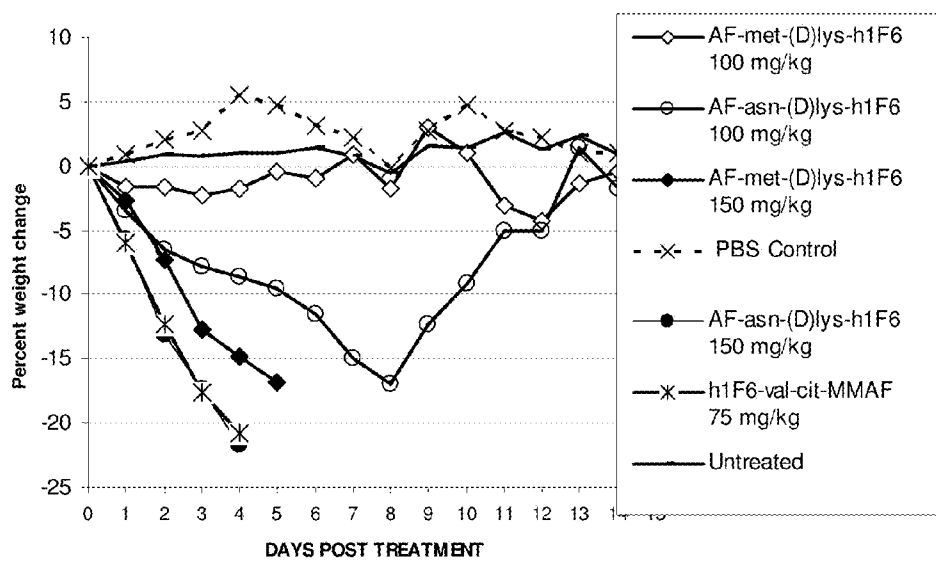


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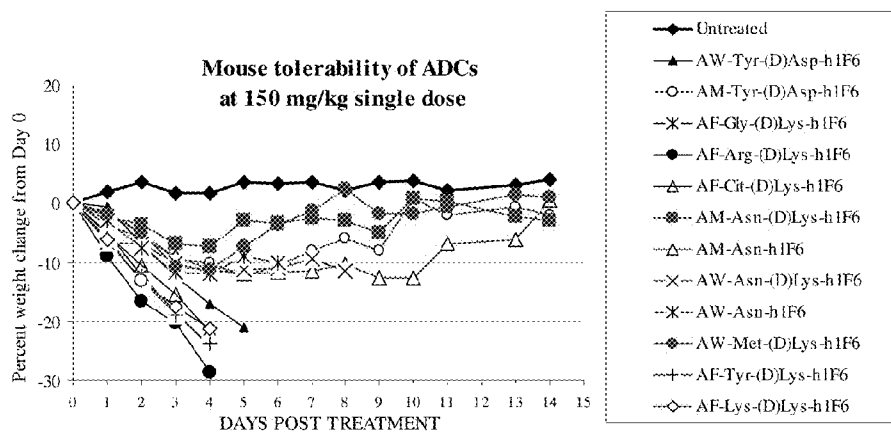


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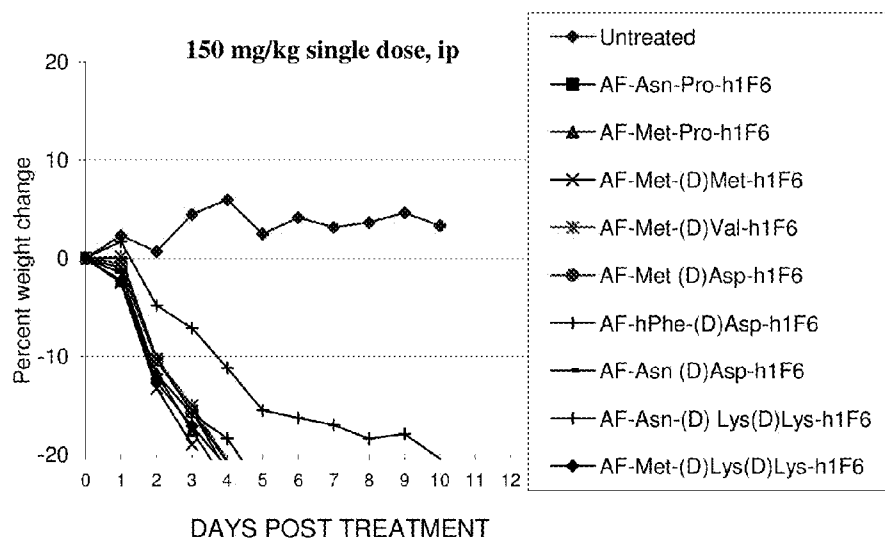


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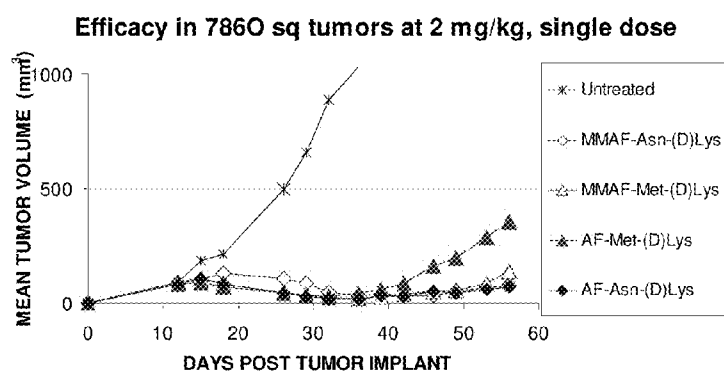


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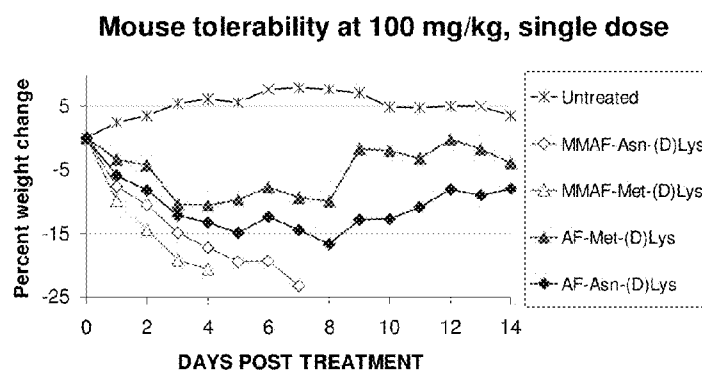


Figure 12

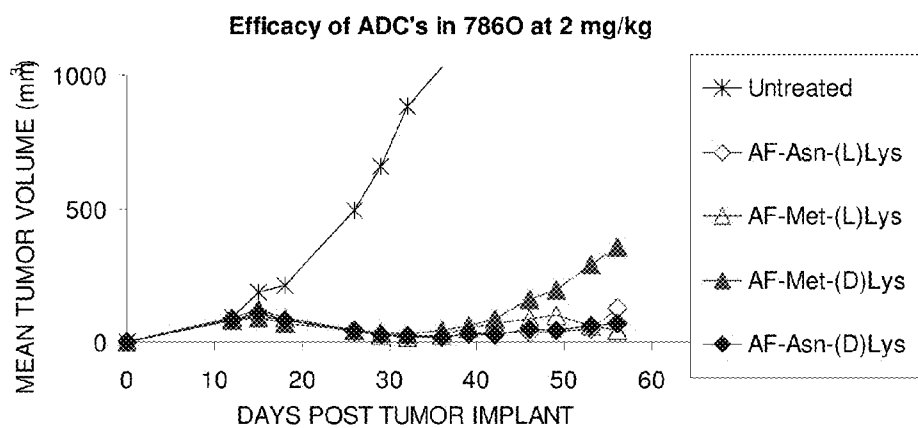


Figure 13

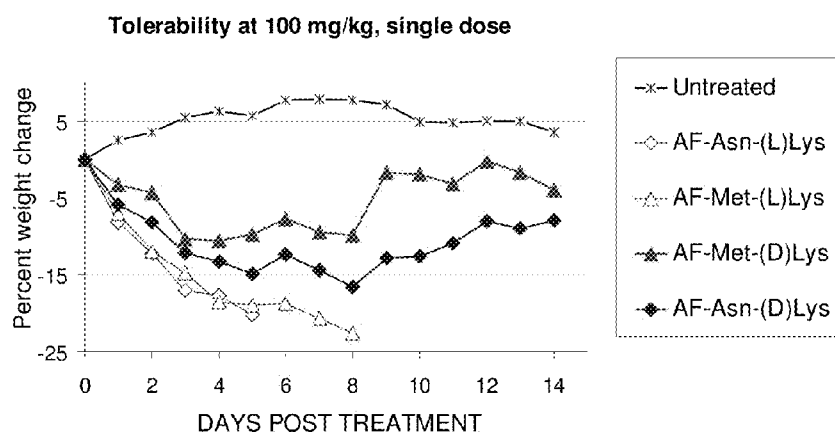


Figure 14

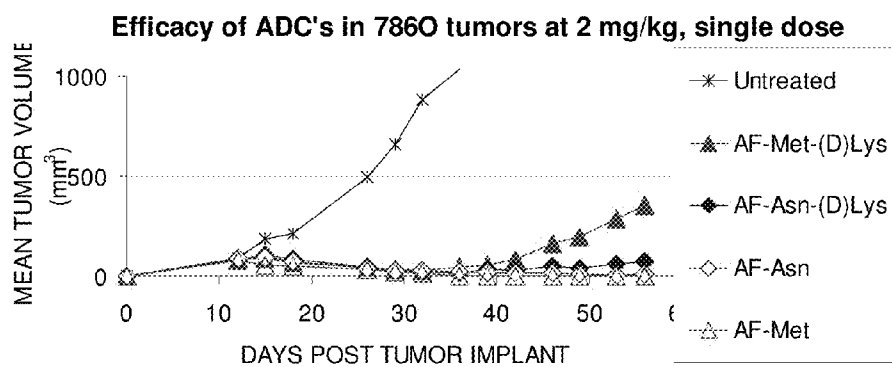


Figure 15

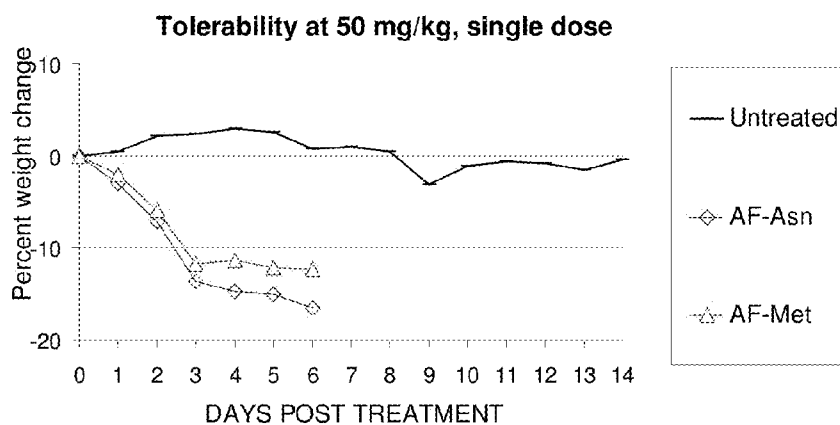


Figure 16

Efficacy of hBU12 ADCs in DoHH-2 tumor xenografts
3 mg/kg, q4dx4

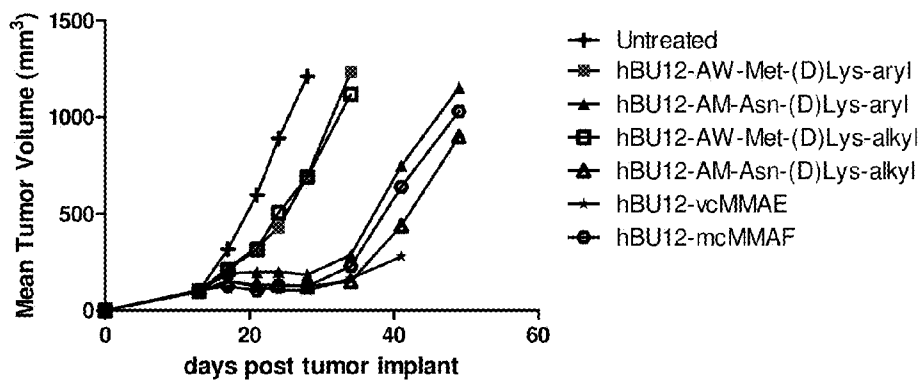
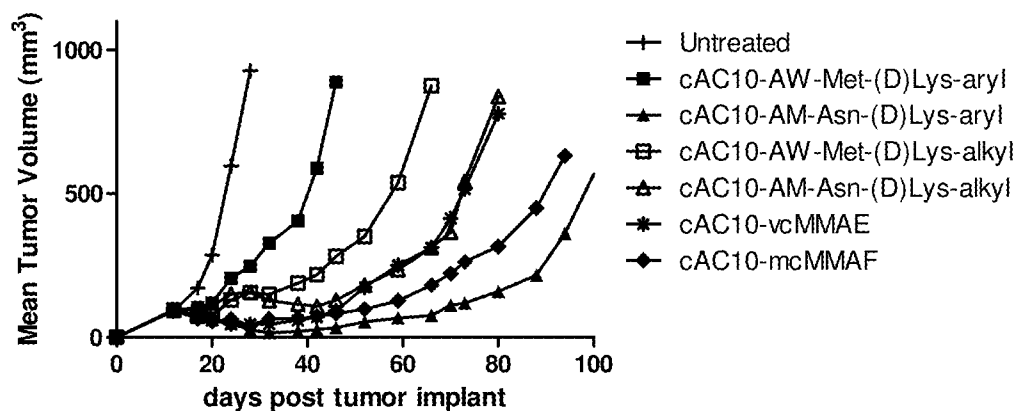


Figure 17

**Efficacy of cAC10 ADCs of maleimide analogues of
AM-Asn-(D)Lys and AW-Met-(D)Lys in L540cy tumor
xenografts
1mg/kg, q4dx3**



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may have more than one charged atom in its structure. Instances where multiple charged atoms are part of the pharmaceutically acceptable salt can have multiple counter ions. Hence, a pharmaceutically acceptable salt can have one or more charged atoms and/or one or more counterion.

"Pharmaceutically acceptable solvate" or "solvate" refer to an association of one or more solvent molecules and a compound of the invention, e.g., a Drug Linker Ligand conjugate or a Drug Linker compound. Examples of solvents that form pharmaceutically acceptable solvates include, but are not limited to, water, isopropanol, ethanol, methanol, DMSO, ethyl acetate, acetic acid, and ethanolamine.

The following abbreviations are used herein and have the indicated definitions: Boc is N-(t-butoxycarbonyl), cit is citrulline, dap is dolaproine, DCM is dichloromethane, DIEA is N,N-diisopropylethylamine, dil is dolaisoleuine, DMF is N,N-dimethylformamide, DMSO is dimethylsulfoxide, doe is dolaphenine, dov is N,N-dimethylvaline, DTNB is 5,5'-dithiobis(2-nitrobenzoic acid), DTPA is diethylenetriaminepentaacetic acid, DTT is dithiothreitol, Fmoc is N-(9-fluorenylmethoxycarbonyl), gly is glycine, HATU is O-(7-azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium hexafluorophosphate, HBTU is 2-[1H-benzotriazole-1-yl]-1,1,3,3-tetramethylaminium hexafluorophosphate; HOBt is 1-hydroxybenzotriazole, HPLC is high pressure liquid chromatography, ile is isoleucine, lys is lysine, MeOH is methanol, MeVal is N-methyl-valine, PAB is p-aminobenzyl, PBS is phosphate-buffered saline (pH 7.4), Ph is phenyl, phe is L-phenylalanine, PyBrop is bromo tris-pyrrolidino phosphonium hexafluorophosphate, TFA is trifluoroacetic acid, UV is ultraviolet, and val is valine.

The following linker abbreviations are used herein and have the indicated definitions: Val Cit or vc is a valine-citrulline dipeptide site in protease cleavable linker; PABC is p-aminobenzylcarbamoyl; (Me)vc is N-methyl-valine citrulline, where the linker peptide bond has been modified to prevent its cleavage by cathepsin B; and MC(PEG)₆-OH is maleimidocaproyl-polyethylene glycol.

The following cytotoxic drug abbreviations are used herein and have the indicated definitions: "Auristatin F" or "AF" is N,N-dimethylvaline-valine-dolaisoleuine(dil)-dolaproine(dap)-phenylalanine. "MMAF" is N-methylvaline-valine-dolaisoleuine(dil)-dolaproine(dap)-phenylalanine (MW 731.5).

Compounds and Conjugates

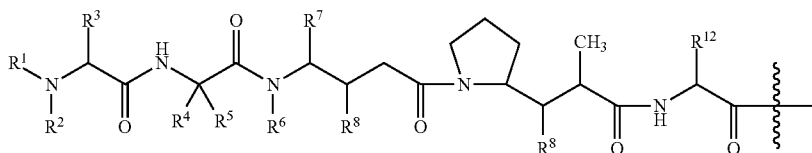
As noted in the Summary of the Invention, the present invention is drawn to a series of compounds and conjugates containing a Drug moiety (D) linked via its C terminus to a Linker unit. The Linker unit can operate to provide a suitable release of D.

In one group of embodiments, the invention provides Drug Linker compounds having Formula I:

LU-D

(I)

or a pharmaceutically acceptable salt or solvate thereof wherein D has the formula:



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wherein the wavy line indicates the attachment to a Linker unit (LU);

R¹ and R² each is independently selected from the group consisting of —H and —C₁-C₈ alkyl, with the proviso that both R¹ and R² are not —H;

R³ is selected from the group consisting of —H, —C₁-C₈ alkyl, —C₃-C₈ carbocycle, aryl, —X¹-aryl, —X¹—(C₃-C₈ carbocycle), —C₃-C₈ heterocycle and —X¹—(C₃-C₈ heterocycle);

R⁴ is selected from the group consisting of —H, —C₁-C₈ alkyl, —C₃-C₈ carbocycle, aryl, —X¹-aryl, —C₁-C₈ alkyl-(C₃-C₈ carbocycle), —C₃-C₈ heterocycle and —X¹—(C₃-C₈ heterocycle);

R⁵ is selected from the group consisting of —H and methyl;

or R⁴ and R⁵ jointly form a carbocyclic ring and have the formula —(CR^aR^b)_n— wherein R^a and R^b are independently selected from the group consisting of —H and —C₁-C₈ alkyl and n is selected from the group consisting of 2, 3, 4, 5 and 6;

R⁶ is selected from the group consisting of —H and —C₁-C₈ alkyl;

R⁷ is selected from the group consisting of —H, —C₁-C₈ alkyl, —C₃-C₈ carbocycle, aryl, —X¹-aryl, —X¹—(C₃-C₈ carbocycle), —C₃-C₈ heterocycle and —X¹—(C₃-C₈ heterocycle);

each R⁸ is independently selected from the group consisting of —H, —OH, —C₁-C₈ alkyl, —C₃-C₈ carbocycle and —O—(C₁-C₈ alkyl);

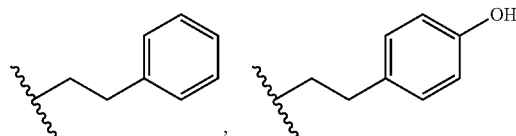
each X¹ is independently —C₁-C₁₀ alkylene-; and

R¹² is selected from the group consisting of —H, —C₁-C₈ alkyl, aryl, —X¹-aryl, —C₃-C₈ carbocycle, —X¹—(C₃-C₈ heterocycle), —C₁-C₈ alkylene-NH₂, —C₃-C₈ heterocycle and —X¹—(C₃-C₈ heterocycle);

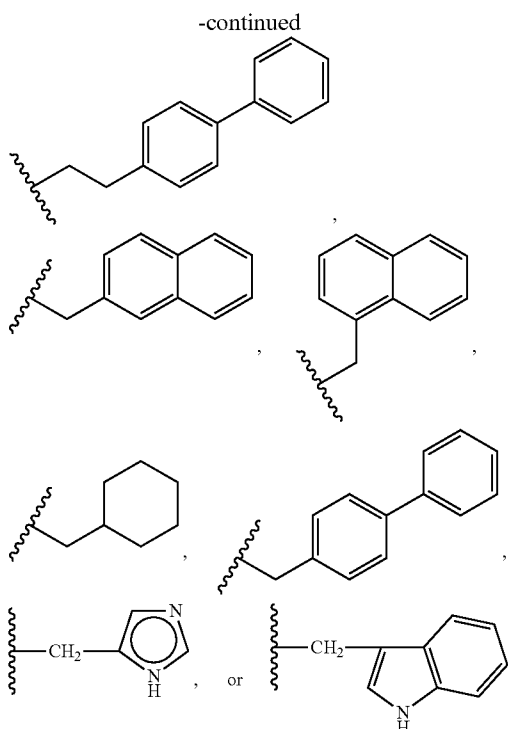
or a pharmaceutically acceptable salt or solvate thereof.

In some embodiments, R¹² is selected from the group consisting of side chains of natural and non-natural amino acids.

In some embodiments, R¹² is selected from the group consisting of H, methyl, isopropyl, isobutyl, sec-butyl, benzyl, p-hydroxybenzyl, —CH₂OH, —CH(OH)CH₃, —CH₂CH₂SCH₃, —CH₂CONH₂, —CH₂COOH, —CH₂CH₂CONH₂, —CH₂CH₂COOH, —(CH₂)₃NHC(=NH)NH₂, —(CH₂)₃NH₂, —(CH₂)₃NHCOCH₃, —(CH₂)₃NHCHO, —(CH₂)₄NHC(=NH)NH₂, —(CH₂)₄NH₂, —(CH₂)₄NHCOCH₃, —(CH₂)₄NHCHO, —(CH₂)₃NHCONH₂, —(CH₂)₄NHCONH₂, —CH₂CH₂CH(OH)CH₂NH₂, 2-pyridylmethyl-, 3-pyridylmethyl-, 4-pyridylmethyl-, phenyl, cyclohexyl,



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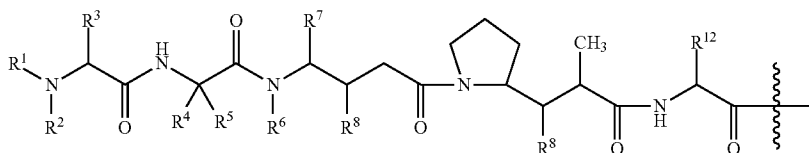


In some embodiments, R^{12} is selected from the group consisting of side chains of natural amino acids. In some embodiments, R^{12} is the side chain of phenylalanine. In some embodiments, R^{12} is the side chain of methionine. In some embodiments, R^{12} is the side chain of tryptophan.

In another group of embodiments, the invention provides Drug Linker compounds having Formula I:



or a pharmaceutically acceptable salt or solvate thereof; wherein D has the formula:



wherein the wavy line indicates the attachment to a Linker unit (LU);

R^1 and R^2 each is independently selected from the group consisting of $-H$ and $-C_1-C_8$ alkyl, with the proviso that both R^1 and R^2 are not $-H$;

R^3 is selected from the group consisting of $-H$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle, aryl, $-C_1-C_8$ alkyl-aryl, $-X^1-(C_3-C_8$ carbocycle), $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle);

R^4 is selected from the group consisting of $-H$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle, aryl, $-X^1$ -aryl, $-C_1-C_8$ alkyl- $(C_3-C_8$ carbocycle), $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle);

R^5 is selected from the group consisting of $-H$ and methyl;

or R^4 and R^5 jointly form a carbocyclic ring and have the formula $-(CR^aR^b)_n-$ wherein R^a and R^b are indepen-

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dently selected from the group consisting of $-H$ and $-C_1-C_8$ alkyl and n is selected from the group consisting of 2, 3, 4, 5 and 6;

R^6 is selected from the group consisting of $-H$ and $-C_1-C_8$ alkyl;

R^7 is selected from the group consisting of $-H$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle, aryl, $-X^1$ -aryl, $-X^1-(C_3-C_8$ carbocycle), $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle);

each R^8 is independently selected from the group consisting of $-H$, $-OH$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle and $-O-(C_1-C_8$ alkyl);

R^{12} is selected from the group consisting of $-H$, $-C_1-C_8$ alkyl, aryl, $-X^1$ -aryl, $-C_3-C_8$ carbocycle, $-X^1-(C_3-C_8$ heterocycle), $-C_1-C_8$ alkylene- NH_2 , $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle); and each X^1 is independently $-C_1-C_{10}$ alkylene-;

the moiety $LU-$ is a Linker unit having the formula $-W_w-A_a-$;

W_w is a sequence of w independently selected amino acid diradicals;

w is an integer ranging from 1 to 12;

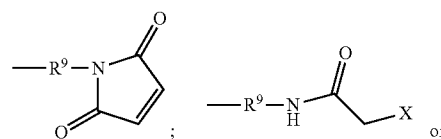
A is a Stretcher unit, and

a is 1 or 2;

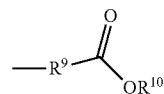
or a pharmaceutically acceptable salt or solvate thereof.

In some embodiments, R^{12} is a side chain of a natural amino acid. In some embodiments, R^{12} is the side chain of phenylalanine. In some embodiments, R^{12} is the side chain of methionine. In some embodiments, R^{12} is the side chain of tryptophan.

In some embodiments, the Stretcher unit A is $-NH-R^9-R^{11}$ or $-O-R^9-R^{11}$, wherein $-R^9-R^{11}$ has the formula:



-continued



wherein R^9 can be selected from the group consisting of $-C_1-C_{10}$ alkylene-, $-C_3-C_8$ carbocycle-, -arylene-, $-C_1-C_{30}$ heteroalkylene-, $-C_3-C_8$ heterocycle-, $-C_1-C_{10}$ alkylene-arylene-, -arylene- $-C_1-C_{10}$ alkylene-, $-C_1-C_{10}$ alkylene- $(C_3-C_8$ carbocycle)-, $-(C_3-C_8$ carbocycle)- $-C_1-C_{10}$ alkylene-, $-C_1-C_{10}$ alkylene- $(C_3-C_8$ heterocycle)-, and $-(C_3-C_8$ heterocycle)- $-C_1-C_{10}$ alkylene-;

wherein X is a leaving group; and

each R^{10} forms an activated ester, wherein R^{10} is independently selected from the group consisting of H ,

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—C₁-C₁₀ alkyl, —C₃-C₈ carbocycle, -aryl, —C₁-C₃₀ heteroalkyl, —C₃-C₈ heterocyclo, —C₁-C₁₀ alkylene-aryl, -arylene-C₁-C₁₀ alkyl, —C₁-C₁₀ alkylene-(C₃-C₈ carbocycle), —(C₃-C₈ carbocycle)-C₁-C₁₀ alkyl, —C₁-C₁₀ alkylene-(C₃-C₈ heterocyclo), and —(C₃-C₈ heterocyclo)-C₁-C₁₀ alkyl.

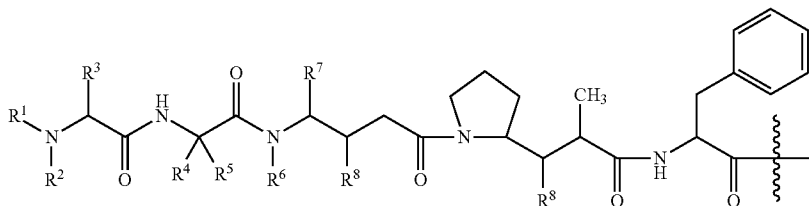
In some embodiments of the Stretcher unit A, —NH—R⁹— is selected from —NH—C₁-C₁₀ alkylene-, —NH—C₁-C₁₀ alkylene-NH—C(O)—C₁-C₁₀ alkylene-, —NH—C₁-C₁₀ alkylene-C(O)—NH—C₁-C₁₀ alkylene-, —NH—(CH₂CH₂O)_r—, —NH—(CH₂CH₂O)_r—CH₂—, —NH—(CH₂CH₂NH)_r—(CH₂)_r—, —NH—(CH₂CH₂NH)—(CH₂)_r—, —NH—C(O)—(CH₂)_r—, —NH—(C₃-C₈ carbocycle)-, —NH-(arylene)-, and —NH—(C₃-C₈ heterocyclo)-, wherein each r is independently 1-10.

In some embodiments of the Stretcher unit A, —O—R⁹— is selected from —O—C₁-C₁₀ alkylene-, —O—C₁-C₁₀ alkylene-NH—C(O)—C₁-C₁₀ alkylene-, —O—C₁-C₁₀ alkylene-C(O)—NH—C₁-C₁₀ alkylene-, —O—(CH₂CH₂O)_r—, —O—(CH₂CH₂O)_r—CH₂—, —O—(C₃-C₈ carbocycle)-, —O-(arylene)-, and —O—(C₃-C₈ heterocyclo)-, wherein each r is independently 1-10.

In embodiments in which the Stretcher unit A is —O—R⁹—R¹¹—, the ester is a hindered ester.

In some embodiments, R⁹ is a polyamine.

In some embodiments, D has the formula

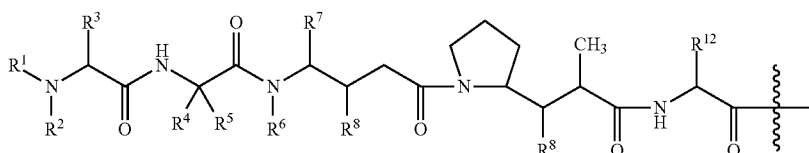


wherein R¹-R⁸ are as set forth above.

In a related aspect, the present invention provides Drug Linker Ligand conjugates in which the Drug Linker compounds further comprise a Ligand unit (L), the conjugates having the formula (II):



or a pharmaceutically acceptable salt or solvate thereof; wherein D has the formula:



wherein the wavy line indicates the attachment to a Linker Unit (LU);

R¹ and R² is independently selected from the group consisting of —H and —C₁-C₈ alkyl, with the proviso that both R¹ and R² are not —H;

R³ is selected from the group consisting of —H, —C₁-C₈ alkyl, —C₃-C₈ carbocycle, aryl, —C₁-C₈ alkyl-aryl, —X¹-(C₃-C₈ carbocycle), —C₃-C₈ heterocycle and —X¹-(C₃-C₈ heterocycle);

R⁴ is selected from the group consisting of —H, —C₁-C₈ alkyl, —C₃-C₈ carbocycle, aryl, —X¹-aryl, —C₁-C₈ alkyl-(C₃-C₈ carbocycle), —C₃-C₈ heterocycle and —X¹-(C₃-C₈ heterocycle);

R⁵ is selected from the group consisting of —H and methyl;

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or R⁴ and R⁵ jointly form a carbocyclic ring and have the formula —(CR^aR^b)_n— wherein R^a and R^b are independently selected from the group consisting of —H and —C₁-C₈ alkyl and n is selected from the group consisting of 2, 3, 4, 5 and 6;

R⁶ is selected from the group consisting of —H and —C₁-C₈ alkyl;

R⁷ is selected from the group consisting of —H, —C₁-C₈ alkyl, —C₃-C₈ carbocycle, aryl, —X¹-aryl, —X¹-(C₃-C₈ carbocycle), —C₃-C₈ heterocycle and —X¹-(C₃-C₈ heterocycle);

each R⁸ is independently selected from the group consisting of —H, —OH, —C₁-C₈ alkyl, —C₃-C₈ carbocycle and —O—(C₁-C₈ alkyl);

each X¹ is independently —C₁-C₁₀ alkylene-;

R¹² is selected from the group consisting of —H, —C₁-C₈ alkyl, aryl, —X¹-aryl, —C₃-C₈ carbocycle, —X¹-(C₃-C₈ heterocycle), —C₁-C₈ alkylene-NH₂, —C₃-C₈ heterocycle and —X¹-(C₃-C₈ heterocycle);

the moiety LU- is a Linker unit having the formula —W_w-A_a-;

W_w is a sequence of w independently selected amino acid diradicals;

w is an integer ranging from 1 to 12;

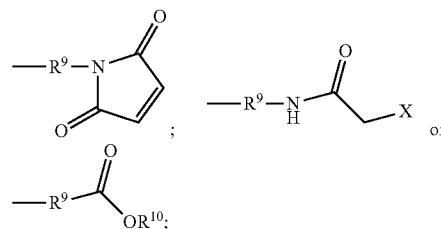
A is a Stretcher unit

a is 1 or 2; and

p is an integer of from 1 to 20;

or a pharmaceutically acceptable salt or solvate thereof.

In some embodiments, the Stretcher unit A is —NH—R⁹—R¹¹ or —O—R⁹—R¹¹, wherein —R⁹—R¹¹ has the formula:



wherein R⁹ can be selected from the group consisting of —C₁-C₁₀alkylene-, —C₃-C₈carbocyclo-, -arylene-, —C₁-C₃₀heteroalkylene-, —C₃-C₈heterocyclo-, —C₁-

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C₁₀alkylene-arylene-, -arylene-C₁-C₁₀alkylene-, —C₁-C₁₀alkylene-(C₃-C₈carbocyclo)-, —(C₃-C₈carbocyclo)-C₁-C₁₀alkylene-, —C₁-C₁₀alkylene-(C₃-C₈heterocyclo)-, and —(C₃-C₈heterocyclo)-C₁-C₁₀alkylene-;

wherein X is a leaving group; and

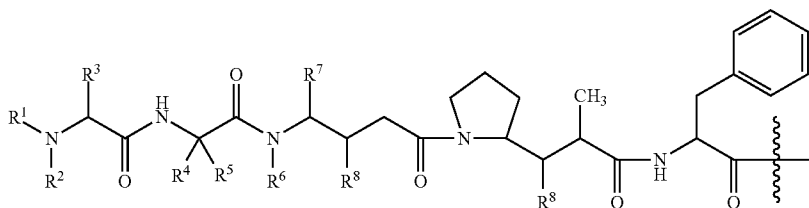
each R¹⁰ forms an activated ester, wherein R¹⁰ is independently selected from the group consisting of —H, —C₁-C₁₀ alkyl, —C₃-C₈ carbocycle, aryl, —C₁-C₃₀ heteroalkyl, —C₃-C₈ heterocyclo, —C₁-C₁₀ alkylene-aryl, -arylene-C₁-C₁₀ alkyl, —C₁-C₁₀ alkylene-(C₃-C₈carbocyclo), —(C₃-C₈carbocyclo)-C₁-C₁₀ alkyl, —C₁-C₁₀ alkylene-(C₃-C₈heterocyclo), and —(C₃-C₈heterocyclo)-C₁-C₁₀ alkyl.

In some embodiments of the Stretcher unit A, —NH—R⁹— is selected from —NH—C₁-C₁₀ alkylene-, —NH—C₁-C₁₀ alkylene-NH—C(O)—C₁-C₁₀ alkylene-, —NH—C₁-C₁₀ alkylene-C(O)—NH—C₁-C₁₀ alkylene-, —NH—(CH₂CH₂O)_r—, —NH—(CH₂CH₂O)_r—CH₂—, —NH—(CH₂CH₂NH)_r—, —NH—(CH₂CH₂NH)_r—(CH₂)_r—NH—C(O)—(CH₂)_r—, —NH—(C₃-C₈carbocyclo)-, —NH-(arylene)-, and —NH—(C₃-C₈heterocyclo)-, wherein each r is independently 1-10.

In some embodiments of the Stretcher unit A, —O—R⁹— is selected from —O—C₁-C₁₀ alkylene-, —O—C₁-C₁₀ alkylene-NH—C(O)—C₁-C₁₀ alkylene-, —O—C₁-C₁₀ alkylene-C(O)—NH—C₁-C₁₀ alkylene-, —O—(CH₂CH₂O)_r—, O—(CH₂CH₂O)_r—CH₂—, —O—(C₃-C₈carbocyclo)-, —O-(arylene)-, and —O—(C₃-C₈heterocyclo)-, wherein each r is independently 1-10.

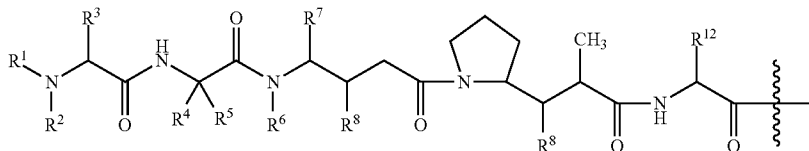
In embodiments in which the Stretcher unit A is —O—R⁹—R¹¹—, the ester is a hindered ester.

In some embodiments of formula I or II, D has the formula

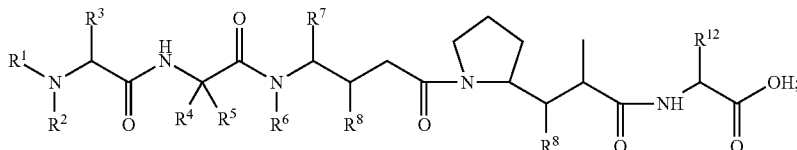


or a pharmaceutically acceptable salt or solvate thereof, wherein R¹-R⁸ are as set forth herein.

Drug Linker Ligand conjugates and Drug Linker compounds also include those wherein D has the formula:



when conjugated; and, as a free drug, D has the formula:



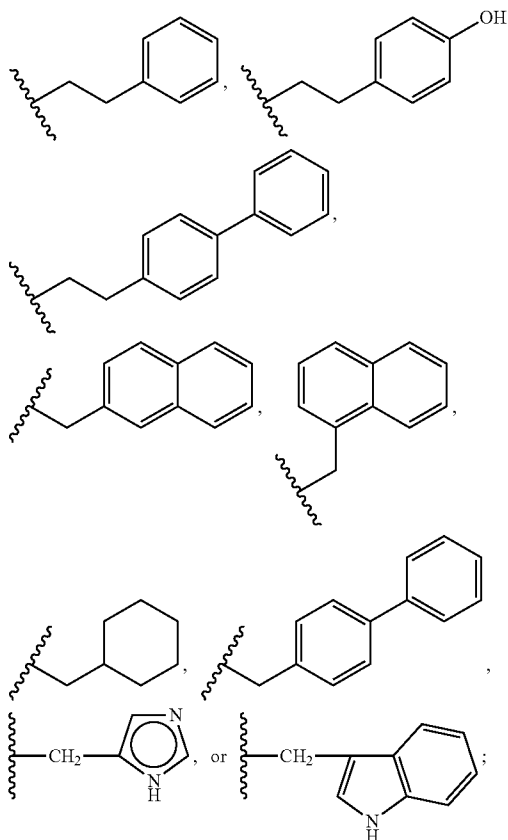
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or a pharmaceutically acceptable salt of solvate thereof, wherein for both formulas:

- R¹ and R² are independently selected from the group consisting of —H and —C₁-C₈ alkyl, with the proviso that both R¹ and R² are not —H;
- R¹ and R² are independently selected from the group consisting of —H and —C₁-C₈ unsubstituted alkyl, with the proviso that both R¹ and R² are not —H;
- R¹ and R² are independently selected from the group consisting of —H and methyl with the proviso that both R¹ and R² are not —H;
- R³ is selected from the group consisting of —H, —C₁-C₈ alkyl, —C₃-C₈ carbocycle, aryl, —X¹-aryl, —X¹-(C₃-C₈carbocycle), —C₃-C₈heterocycle and —X¹-(C₃-C₈heterocycle);
- R³ is selected from the group consisting of —H or —C₁-C₈ alkyl;
- R³ is selected from the group consisting of —H or unsubstituted —C₁-C₈ alkyl;
- R³ is selected from the group consisting of —H or isopropyl;
- R⁴ is selected from the group consisting of —H, —C₁-C₈ alkyl, —C₃-C₈ carbocycle, aryl, —X¹-aryl, —C₁-C₈ alkyl-(C₃-C₈carbocycle), —C₃-C₈heterocycle and —X¹-(C₃-C₈heterocycle);
- R⁴ is a side chain of a natural amino acid;
- R⁴ is —H, methyl, isopropyl, isobutyl, sec-butyl, benzyl, p-hydroxybenzyl, —CH₂OH, —CH(OH)CH₃, —CH₂CH₂SCH₃, —CH₂CONH₂, —CH₂COOH, —CH₂CH₂CONH₂, —CH₂CH₂COOH, —(CH₂)₃NHC(=NH)NH₂, —(CH₂)₃NH₂, —(CH₂)₃NHCOCH₃, —(CH₂)₃NHCHO, —(CH₂)₄NHC(=NH)NH₂, —(CH₂)₄NH₂, —(CH₂)₄NHCOCH₃, —(CH₂)₄

NHCHO, —(CH₂)₃NHCONH₂, —(CH₂)₄NHCONH₂, —CH₂CH₂CH(OH)CH₂NH₂, 2-pyridylmethyl-, 3-pyridylmethyl-, 4-pyridylmethyl-, phenyl, cyclohexyl,

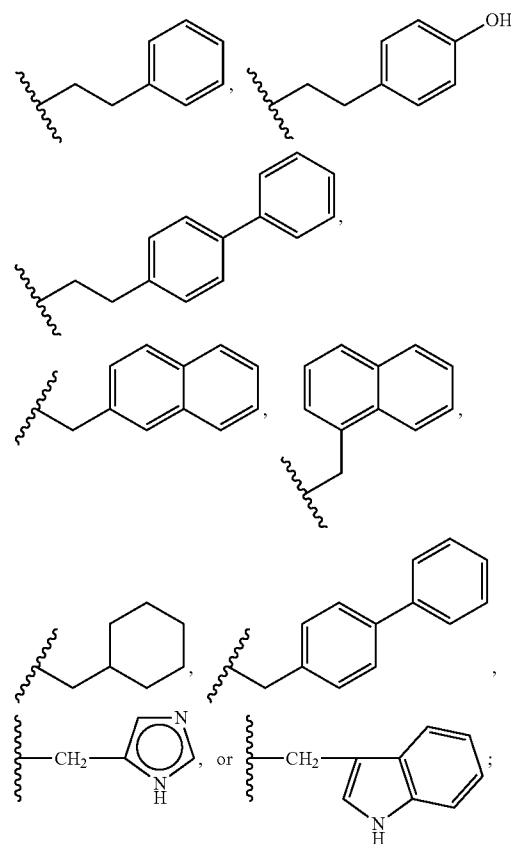
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- (k) R^4 is isopropyl and R^5 is —H;
 (l) or R^4 and R^5 jointly form a carbocyclic ring and have the formula $-(CR^aR^b)_n$ wherein R^a and R^b are independently selected from the group consisting of —H and $-C_1-C_8$ alkyl and n is selected from the group consisting of 2, 3, 4, 5 and 6;
 (m) R^6 is selected from the group consisting of —H and $-C_1-C_8$ alkyl;
 (n) R^6 is selected from the group consisting of —H and unsubstituted $-C_1-C_8$ alkyl;
 (o) R^6 is selected from the group consisting of —H and methyl;
 (p) R^7 is selected from the group consisting of —H, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle, aryl, $-X^1$ -aryl, $-X^1-(C_3-C_8$ carbocycle), $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle);
 (q) R^7 is selected from the group consisting of —H and $-C_1-C_8$ alkyl;
 (r) R^7 is selected from the group consisting of —H and unsubstituted $-C_1-C_8$ alkyl;
 (s) R^7 is selected from the group consisting of —H and sec-butyl;
 (t) each R^8 is independently selected from the group consisting of —H, —OH, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle and $-O-(C_1-C_8$ alkyl);
 (u) each R^8 is independently selected from the group consisting of —H and $-O-(C_1-C_8$ alkyl);
 (v) each R^8 is independently selected from the group consisting of —H and unsubstituted $-O-(C_1-C_8$ alkyl);
 (w) each R^8 is independently selected from the group consisting of —H and unsubstituted $-OCH_3$;

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- (x) each X^1 is independently $-C_1-C_{10}$ alkylene-;
 (y) each X^1 is independently unsubstituted $-C_1-C_{10}$ alkylene-;
 (z) R^{12} is selected from the group consisting of —H, $-C_1-C_8$ alkyl, aryl, $-X^1$ aryl, $-C_3-C_8$ carbocycle, $-X^1-(C_3-C_8$ heterocycle), $-C_1-C_8$ alkylene- NH_2 , $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle);
 (aa) R^{12} is a side chain of a natural amino acid;
 (bb) R^{12} is H, methyl, isopropyl, isobutyl, sec-butyl, benzyl, p-hydroxybenzyl, $-CH_2OH$, $-CH(OH)CH_3$, $-CH_2CH_2SCH_3$, $-CH_2CONH_2$, $-CH_2COOH$, $-CH_2CH_2CONH_2$, $-CH_2CH_2COOH$, $-(CH_2)_3NHC(=NH)NH_2$, $-(CH_2)_3NH_2$, $-(CH_2)_3NHCOCH_3$, $-(CH_2)_3NHCHO$, $-(CH_2)_4NHC(=NH)NH_2$, $-(CH_2)_4NH_2$, $-(CH_2)_4NHCOCH_3$, $-(CH_2)_4NHCHO$, $-(CH_2)_3NHCONH_2$, $-(CH_2)_4NHCONH_2$, $-CH_2CH_2CH(OH)CH_2NH_2$, 2-pyridylmethyl-, 3-pyridylmethyl-, 4-pyridylmethyl-, phenyl, cyclohexyl,



and any combination of embodiments (a) to (bb), provided that it is understood that combinations in which different embodiments of the same substituent would be combined are excluded. In an exemplary embodiment, the alkyl groups in groups (a) to (bb) are unsaturated.

In exemplary embodiments, R^5 is —H, and R^4 and R^{12} independently a side chain of a natural amino acid and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^3 is —H or $-C_1-C_8$ alkyl, R^4 is a side chain of a natural amino acid, R^7 is —H or $-C_1-C_8$ alkyl, and R^8 is $-O-(C_1-C_8$ alkyl) and the remaining groups are as indicated in any of the embodiments herein.

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In exemplary embodiments, R^3 is —H or — C_1 - C_8 alkyl, R^4 is — $CH_2CH_2SCH_3$, — CH_2CONH_2 , — CH_2COOH , — $CH_2CH_2CONH_2$, — CH_2CH_2COOH , — $(CH_2)_3NHC(=NH)NH_2$, — $(CH_2)_3NH_2$, — $(CH_2)_3NHCOCH_3$, — $(CH_2)_3NHCHO$, — $(CH_2)_4NHC(=NH)NH_2$, — $(CH_2)_4NH_2$, — $(CH_2)_4NHCOCH_3$, — $(CH_2)_4NHCHO$, — $(CH_2)_3NHCONH_2$, — $(CH_2)_4NHCONH_2$, — $CH_2CH_2CH(OH)CH_2NH_2$, R^7 is —H or — C_1 - C_8 alkyl, and R^8 is —O—(C_1 - C_8 alkyl) and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^3 is —H or — C_1 - C_8 alkyl, R^4 is — $CH_2CH_2SCH_3$, — CH_2CONH_2 , — CH_2COOH , — $CH_2CH_2CONH_2$, — CH_2CH_2COOH , — $(CH_2)_3NHC(=NH)NH_2$, — $(CH_2)_3NH_2$, — $(CH_2)_3NHCOCH_3$, — $(CH_2)_3NHCHO$, — $(CH_2)_4NHC(=NH)NH_2$, — $(CH_2)_4NH_2$, — $(CH_2)_4NHCOCH_3$, — $(CH_2)_4NHCHO$, — $(CH_2)_3NHCONH_2$, — $(CH_2)_4NHCONH_2$, — $CH_2CH_2CH(OH)CH_2NH_2$, R^7 is —H or — C_1 - C_8 alkyl, R^8 is —O—(C_1 - C_8 alkyl), R^{12} is a side chain of a natural amino acid, and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^3 is —H or — C_1 - C_8 alkyl, R^4 is — $CH_2CH_2SCH_3$, — CH_2CONH_2 , — CH_2COOH , — $CH_2CH_2CONH_2$, — CH_2CH_2COOH , — $(CH_2)_3NHC(=NH)NH_2$, — $(CH_2)_3NH_2$, — $(CH_2)_3NHCOCH_3$, — $(CH_2)_3NHCHO$, — $(CH_2)_4NHC(=NH)NH_2$, — $(CH_2)_4NH_2$, — $(CH_2)_4NHCOCH_3$, — $(CH_2)_4NHCHO$, — $(CH_2)_3NHCONH_2$, — $(CH_2)_4NHCONH_2$, — $CH_2CH_2CH(OH)CH_2NH_2$, R^7 is —H or — C_1 - C_8 alkyl, R^8 is —O—(C_1 - C_8 alkyl), R^{12} is the side chain of phenylalanine, methionine or tryptophan, and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^3 is —H or — C_1 - C_8 alkyl, R^4 is — $CH_2CH_2SCH_3$ or — $CH_2CH(CH_3)CH_3$, R^7 is —H or — C_1 - C_8 alkyl, and R^8 is —O—(C_1 - C_8 alkyl) and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^3 is —H or — C_1 - C_8 alkyl, R^4 is — $CH_2CH_2SCH_3$ or — $CH_2CH(CH_3)CH_3$, R^7 is —H or — C_1 - C_8 alkyl, R^8 is —O—(C_1 - C_8 alkyl), R^{12} is a side chain of a natural amino acid, and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^3 is —H or — C_1 - C_8 alkyl, R^4 is — $CH_2CH_2SCH_3$ or — $CH_2CH(CH_3)CH_3$, R^7 is —H or — C_1 - C_8 alkyl, R^8 is —O—(C_1 - C_8 alkyl), R^{12} is the side chain of phenylalanine, methionine or tryptophan, and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^3 is —H or — C_1 - C_8 saturated alkyl, R^4 is a side chain of a natural amino acid, R^6 is —H or — C_1 - C_8 saturated alkyl; R^7 is —H or C_1 - C_8 saturated alkyl, and R^8 is —O—(C_1 - C_8 saturated alkyl) and the remaining groups are as indicated in any of the embodiments herein.

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In exemplary embodiments, R^3 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^4 is a side chain of a natural amino acid, R^6 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^7 is —H or — C_1 - C_8 unsubstituted saturated alkyl, and R^8 is —O—(C_1 - C_8 unsubstituted saturated alkyl) and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^1 and R^2 are —H or — C_1 - C_8 unsubstituted saturated alkyl with the proviso that both are not hydrogen, R^3 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^4 is a side chain of a natural amino acid, R^6 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^7 is —H or — C_1 - C_8 unsubstituted saturated alkyl, and R^8 is —O—(C_1 - C_8 unsubstituted saturated alkyl) and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^1 and R^2 are —H or — C_1 - C_8 unsubstituted saturated alkyl with the proviso that both are not hydrogen, R^3 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^4 is a side chain of a natural amino acid, R^6 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^7 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^8 is —O—(C_1 - C_8 unsubstituted saturated alkyl); R^{12} is a side chain of a natural amino acid and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^1 and R^2 are —H or — C_1 - C_8 unsubstituted saturated alkyl with the proviso that both are not hydrogen, R^3 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^4 is — $CH_2CH_2SCH_3$ or — $CH_2CH(CH_3)CH_3$, R^6 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^7 is —H or — C_1 - C_8 unsubstituted saturated alkyl, R^8 is —O—(C_1 - C_8 unsubstituted saturated alkyl); R^{12} is the side chain of phenylalanine, methionine or tryptophan and the remaining groups are as indicated in any of the embodiments herein.

In exemplary embodiments, R^1 and R^2 are —H or — C_1 - C_3 unsubstituted saturated alkyl with the proviso that both are not hydrogen, R^3 is — C_1 - C_4 unsubstituted saturated alkyl, R^4 is — $CH_2CH_2SCH_3$ or — $CH_2CH(CH_3)CH_3$, R^5 is —H, R^6 is methyl, R^7 is or — C_1 - C_4 unsubstituted saturated alkyl, R^8 is —O—(C_1 - C_3 unsubstituted saturated alkyl); R^{12} is the side chain of phenylalanine, methionine or tryptophan and the remaining groups are as indicated in any of the embodiments herein.

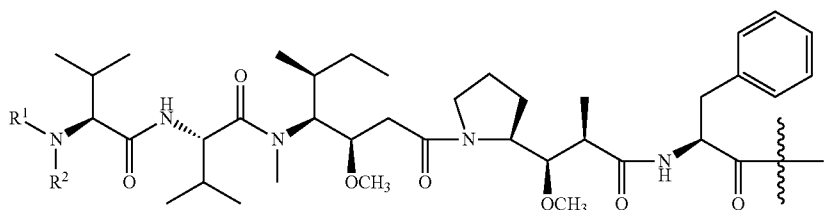
In some embodiments, R^3 , R^4 and R^7 are independently isopropyl or sec-butyl and R^5 is —H. In an exemplary embodiment, R^3 and R^4 are each isopropyl, R^5 is —H, and R^7 is sec-butyl.

In some embodiments, R^2 and R^6 are each —H or — C_1 - C_8 alkyl. In another embodiment, R^2 and R^6 are each — CH_3 , and R^5 is —H.

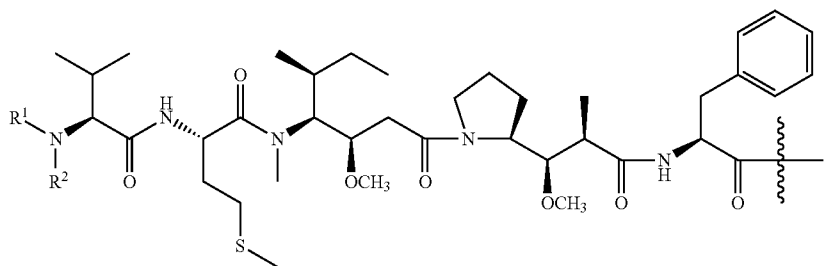
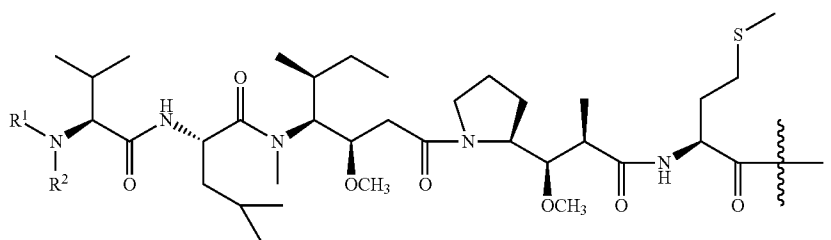
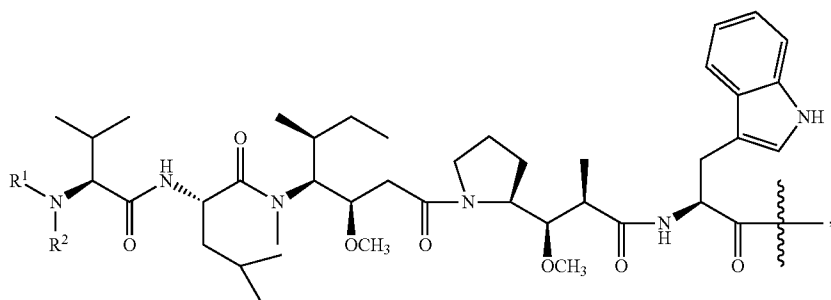
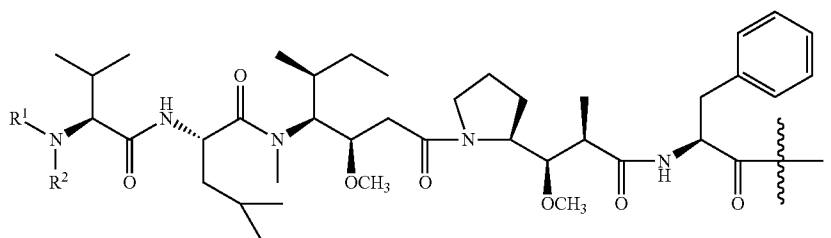
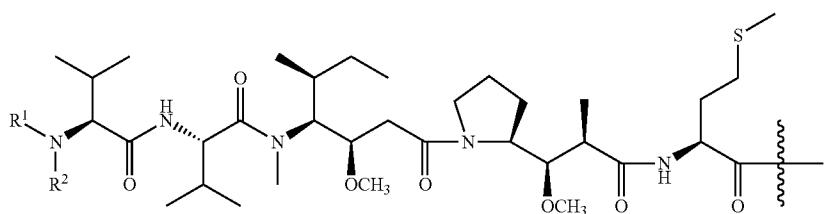
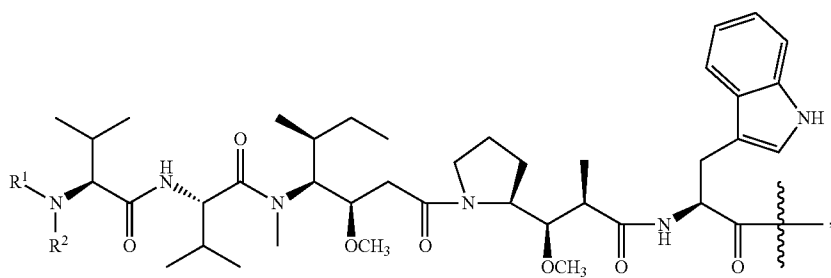
In some embodiments, each occurrence of R^8 is — OCH_3 .

In an exemplary embodiment, R^3 and R^4 are each isopropyl, R^2 and R^6 are each methyl, R^5 is —H, R^7 is sec-butyl, and each occurrence of R^8 is — OCH_3 .

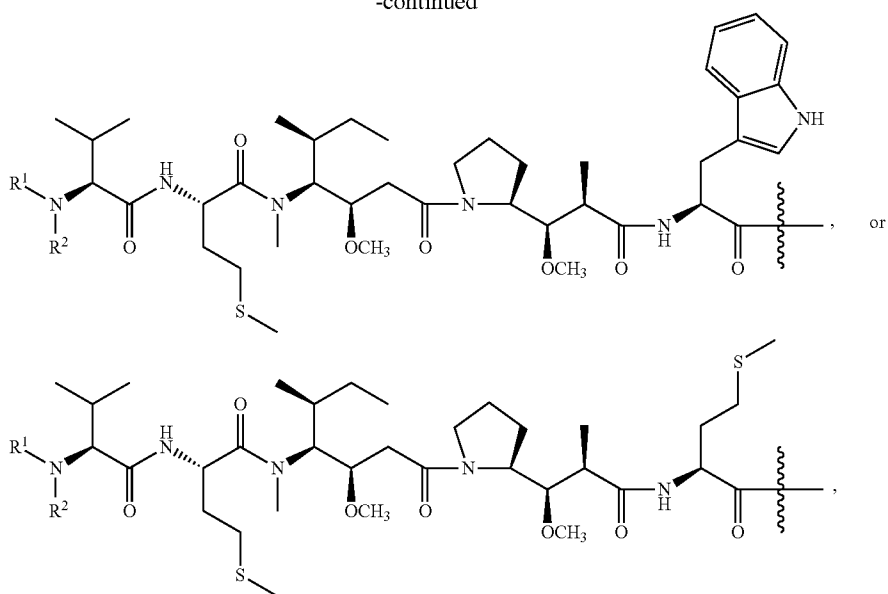
In another aspect of the invention the compound D has the following formula:



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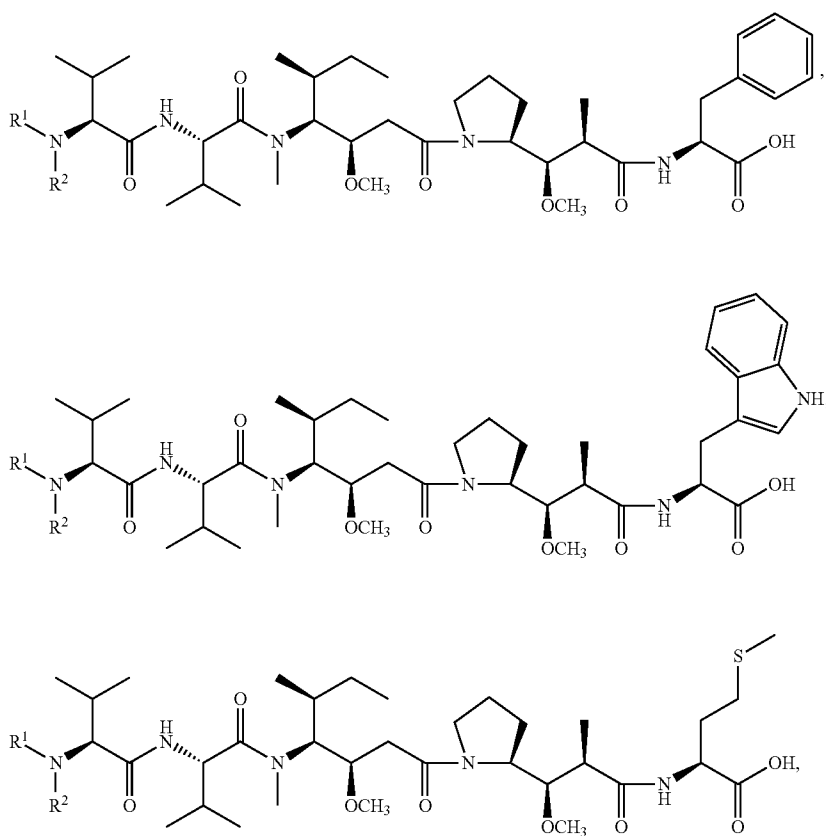


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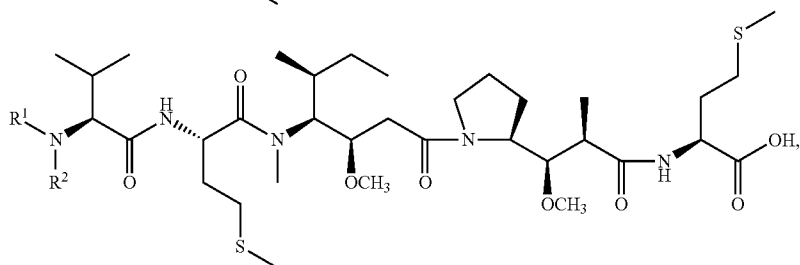
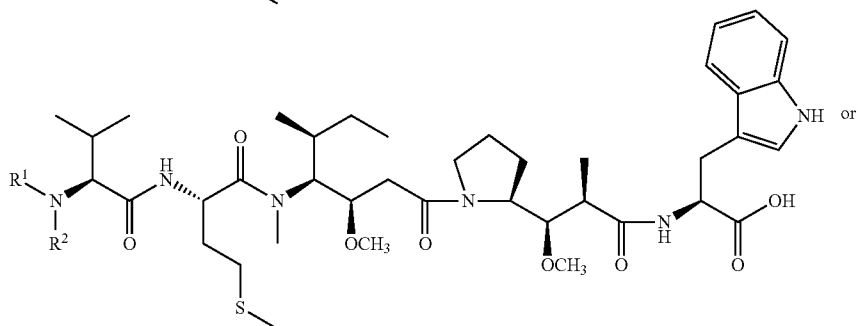
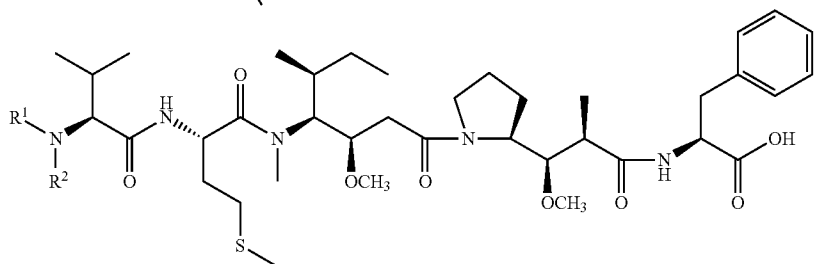
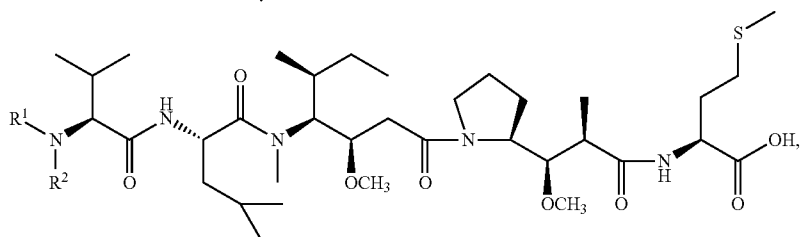
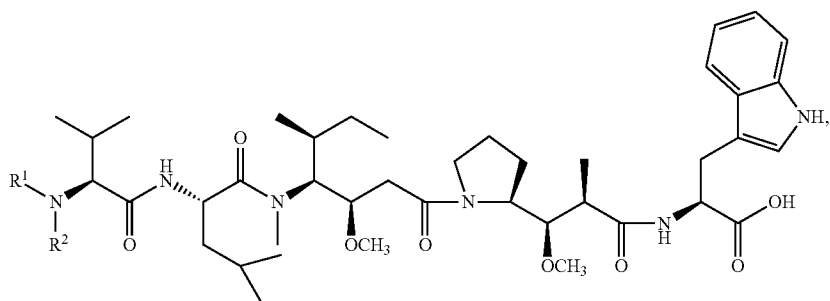
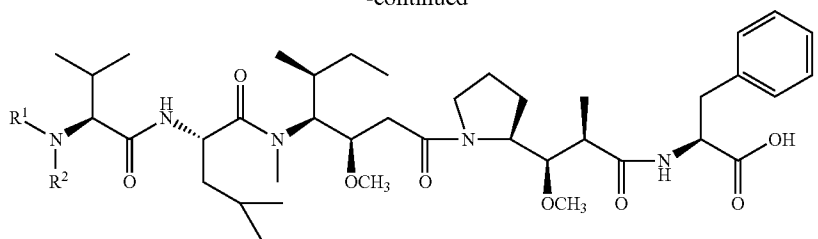


or a pharmaceutically acceptable salt or solvate thereof, wherein the wavy line indicates a covalent bond to the Linker unit (LU).

In another aspect of the invention the compound D has one of the following formulas as a free drug:

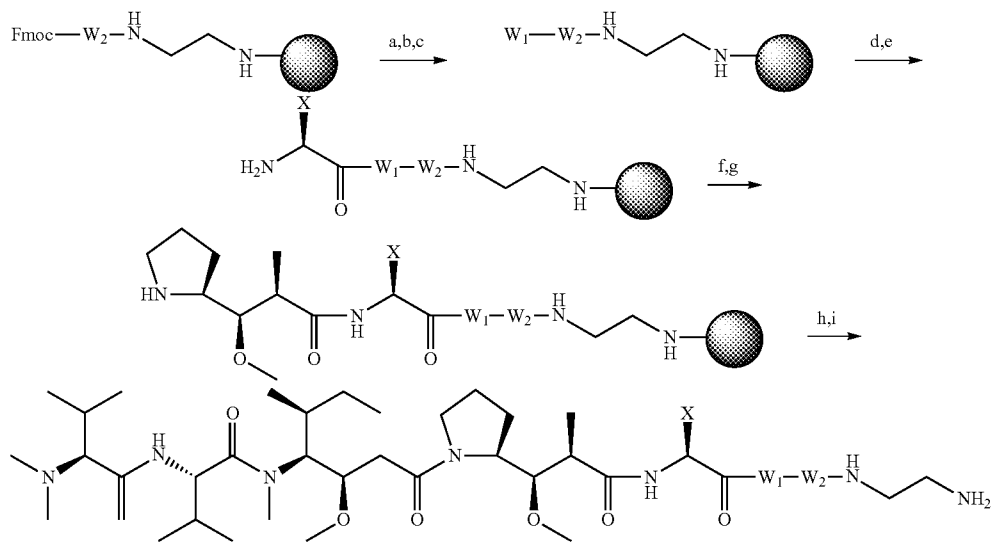


-continued



or a pharmaceutically acceptable salt or solvate thereof.

Scheme 1. Solid Phase Synthesis Route



Reaction conditions (a): 20% piperidine/DMF, (b) Fmoc—W₁ (2-5 equiv.), HATU (2-5 equiv.) and DIEA, (4-10 equiv.), (c) 20% piperidine/DMF, (d) Fmoc—NH—CH(X)—COOH (2 equiv.) HATU (2 equiv.) and DIEA (4 equiv.), (e) 20% piperidine/DMF, (f) Fmoc Dap (2 equiv.) HATU (2 equiv.) and DIEA (4 equiv.), (g) 20% piperidine/DMF, (h) Dov-Val-Dil-OH (2 equiv.), HATU (2 equiv.), and DIEA (4 equiv.), (i): 95% TFA/dichloromethane.

Amino acids not commercially available pre-loaded on an appropriate acid labile resin, preferably 1,2-diaminoethane trityl resin, can be loaded onto 1,2-diaminoethane trityl resin as described in General Procedure SP(a). Loading can be determined by spectrophotometric Fmoc-quantitation assay. Loading levels (mmol/g) of commercially available pre-loaded amino acids on 1,2-diaminoethane trityl resin can be determined as described in General Procedure SP(b). Peptides can then be assembled on the resin loaded with the amino acids of the Amino Acid unit by coupling Fmoc-amino acid using appropriate coupling reagent, preferably HATU/DIEA, followed by Fmoc deprotection and subsequent coupling with reagents which form the Drug unit. Drug unit synthesis can be accomplished by then coupling Fmoc-Dolaproine using appropriate coupling agent, preferably HATU/DIEA, followed by Fmoc deprotection and subsequent coupling of Fmoc-MeVal-Val-Dil tripeptide. Solid phase coupling routine is well established in the art and is described in General Procedure SP(c). Final deprotection of peptides and cleavage off resin can be readily performed following General Procedure SP(d).

Drug Linker compounds containing only one amino acid in the Linker unit were prepared following the above procedure but skipping steps (b) and (c).

Drug Linker compounds containing more than 2 amino acids in the Linker unit were prepared by incorporating additional steps (b), (c) with Fmoc-W₃.

Monomethyl versions of Drug Linkers were prepared by substituting in step (h) Dov-Val-Dil-OH with Fmoc-MeVal-Val-Dil-OH. Fmoc was removed by 20% piperidine/DMF treatment prior to cleaving compound off the resin, step (i).

Auristatins containing various amino acids at the C-terminus of the drug were generated using corresponding Fmoc-NH—CH(X)—COOH in step (d): for example, Fmoc-Phenylalanine for auristatin F (AF); Fmoc-Methionine for auristatin M (AM); Fmoc-Tryptophan for auristatin W (AW), etc.

General Procedure SP(a). Resin Loading

Fmoc-amino acid is suspended in anhydrous solvent such as CH₂Cl₂ and DIEA. The resulting mixture is added to a syringe containing 1,2-diaminoethane trityl resin. The mixture is agitated at room temperature, then the resin is filtered, washed with solvents, such as DCM/MeOH/DIEA, MeOH, DCM, DMF, ethyl ether, and is dried in vacuo. The resin is then left under vacuum overnight.

Loading is determined by Fmoc-quantitation. A known quantity of resin is weighed in a volumetric flask and to the flask is added a solvent such as 20% piperidine/DMF. The mixture is allowed to cleave for about 1 h, with occasional agitation. To the flask is transferred a solvent such as DMF to bring the total volume to a set level (e.g., 10 mL). A blank solution is prepared with an equivalent volume of 20% piperidine/DMF in a volumetric flask. The spectrophotometer is zeroed with the blank solution. The absorbance is measured at 301 nm and the loading level is given by:

$$\text{Loading (mmol/g)} = A_{301} \times 10 \text{ mL} / 7800 \times \text{wt}$$

whereby A₃₀₁ is the absorbance at 301 nm, 7800 is the extinction coefficient of the piperidine-fluorenone adduct, and wt is the weight of resin used in milligrams. Fmoc quantitation is generally performed in duplicate.

General Procedure SP(b). Fmoc Quantitation of Commercially Available Pre-Loaded Resins

Fmoc-Cl is dissolved in anhydrous solvent, such as CH₂Cl₂ to make a working solution. This solution is transferred to a plastic syringe containing amino acid-1,2-diaminoethane trityl resin. The mixture is agitated for about 2 h. The resin is then filtered and washed with appropriate solvents, such as DMF, CH₂Cl₂ or ethyl ether, and dried in-vacuo for about 2 h. The resin is subjected to Kaiser amine test. Upon negative results (free amine fully protected) the Fmoc quantitation to obtain loading level is performed as shown in General Procedure SP(a).

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General Procedure SP(c). Solid Phase Peptide Coupling Using HATU.

A 20% piperidine in DMF solution is added to the syringe with a PET frit containing resin, and the mixture is agitated for about 2 h. The resin is then filtered, washed with appropriate solvents such as DMF, DCM or ethyl ether, and is dried in-vacuo for about 2 h.

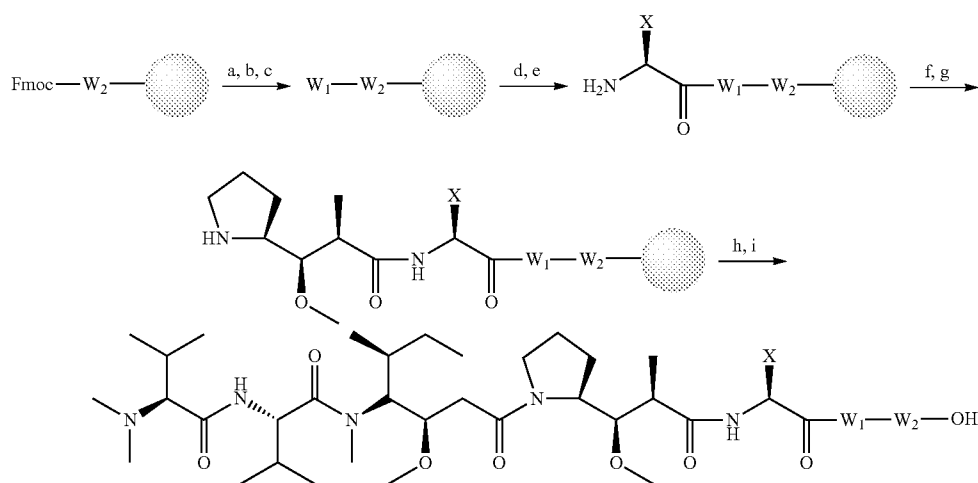
A solution of Fmoc-amino acid, HATU and DIEA in DMF was then added to the resin which was shaken for about 4 hrs, washed, and dried. In this manner, Fmoc-Phe, Fmoc-Dap, and Dov-Val DiI-OH (or Fmoc-MeVal-Val-DiI) were sequentially coupled on the resin as described previously.

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General Procedure SP(d). Final Deprotection and Cleavage Off Resin.

The resin was then treated with a solution of 95% TFA/dichloromethane, and washed with 95% TFA/dichloromethane. The combined filtrate was allowed to stand at room temperature for about 30 min, then concentrated to dryness. The product was purified by reverse phase preparative HPLC.

Alternatively, Drug Linkers can be prepared on chlorotrityl resin as shown in Scheme 1a:



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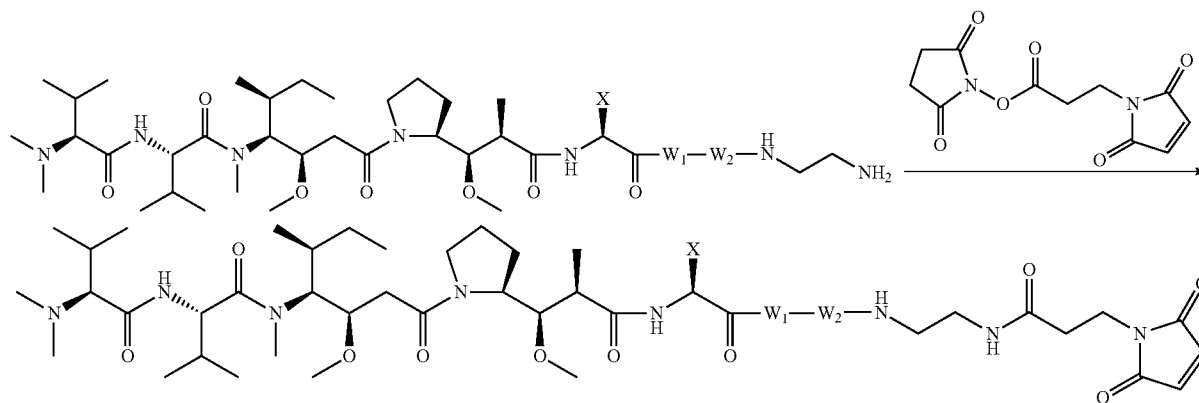
The same reaction conditions are used as in Scheme 1. (Note: the free hydroxyl after steps h, i is part of W_2 .)

Completion of Drug Linker Synthesis

To complete the synthesis of the Drug Linker compound, and to introduce a reactive site that will be used for conjugation with a ligand, the partial Drug Linker compound, as cleaved off the resin, is reacted with another portion of a Stretcher unit to form a complete Drug Linker compound.

In one aspect, the partial Drug Linker compound reacts with a carboxylic acid or an activated ester of the second portion of the Stretcher unit. See, for example Scheme 2:

Scheme 2

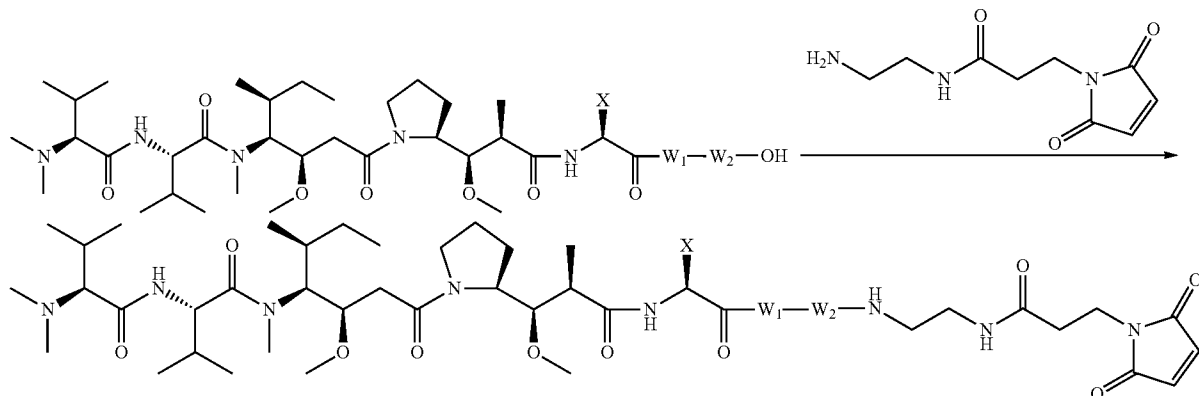


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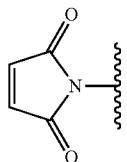
Reaction conditions: Maleimidopropionic acid NHS ester (2 equiv.), DIEA (2 equiv.), DMF.

In another aspect, the partial Drug Linker compound 5 prepared according to Scheme 1a reacts with an amine of the second portion of the Stretcher unit following standard coupling procedures of peptide chemistry. See for example Scheme 2a.

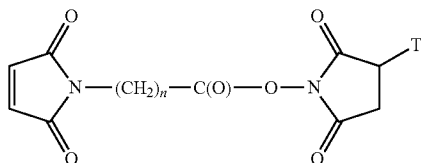
Scheme 2a



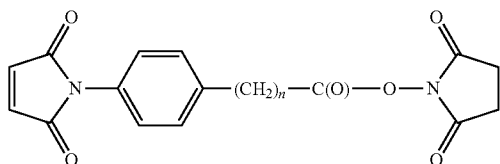
In one aspect, the second portion of the Stretcher unit also contains a thiol-accepting group. Suitable thiol-accepting groups include, for example, haloacetamide groups or maleimide groups having the formula



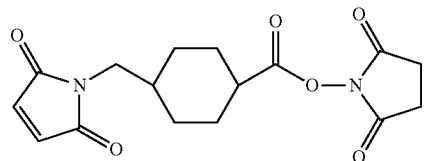
Useful Stretcher units can be obtained via commercial sources, such as Molecular Biosciences Inc. (Boulder, Colo.), or as shown below. In addition, Stretcher units can be prepared as summarized in Scheme 3 below. Scheme 3 illustrates a general synthesis of an illustrative Stretcher unit containing a maleimide Stretcher unit.



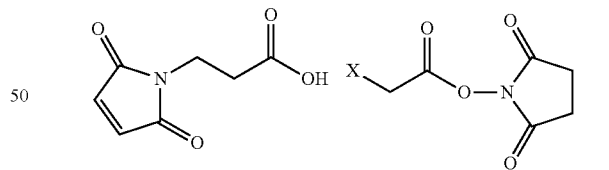
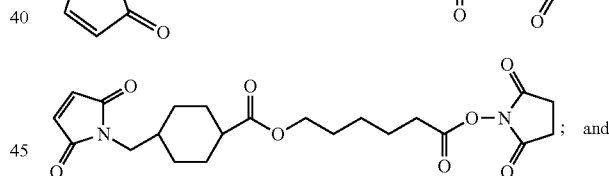
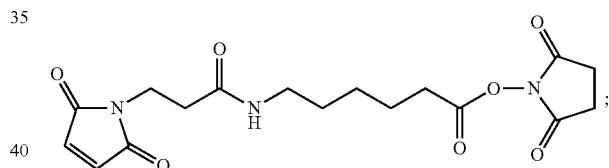
where n is an integer ranging from 1-10 and T is —H or —SO₃Na;

**64**

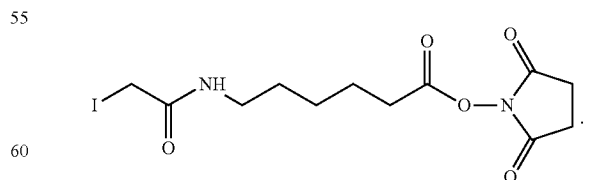
where n is an integer ranging from 0-3;



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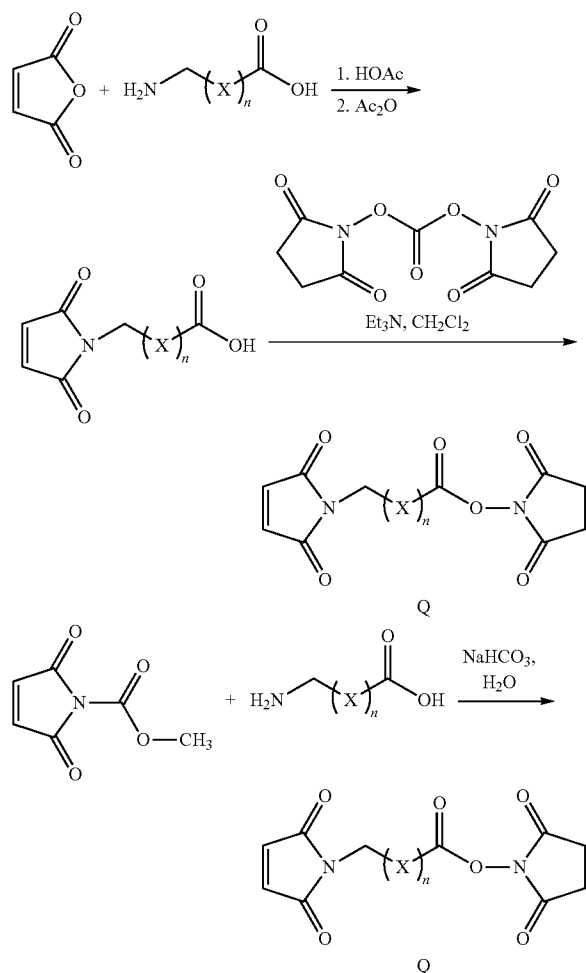
where X is —Br or —I; and



The following Scheme 3 shows methods for obtaining of a Stretcher unit containing an activated ester to react with an Amino Acid unit and having a maleimide for conjugation with a Ligand unit.

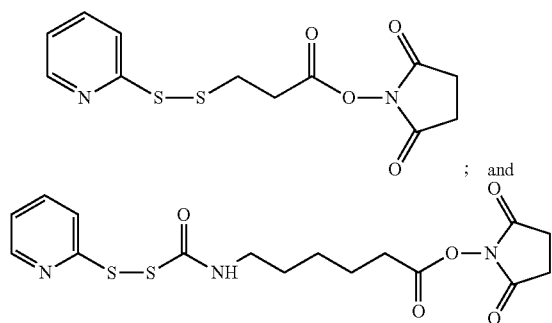
65

Scheme 3



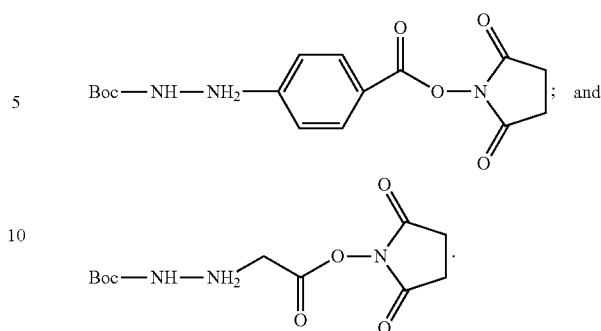
wherein X is -alkylene-, -arylene-, C₃-C₈ carbocyclo-, C₃-C₈-heterocyclo-, -heteroalkylene-; and n is an integer ranging either from 0-1.

Another useful Ligand reactive Stretcher units contains activated disulfides, as shown below. They can be introduced into a Drug Linker compound by reacting the following intermediates with the amine function of the Drug Linker compound:

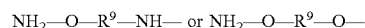


Stretcher units of formula (Va) can be introduced into a Linker unit by reacting the following intermediates with the N-terminus of an Amino Acid unit:

66

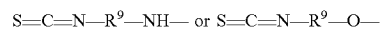


Other useful Stretchers may be synthesized according to known procedures. Aminoxy Stretchers of the formula shown below can be prepared by treating alkyl halides with N-Boc-hydroxylamine according to procedures described in Jones et al., 2000, *Tetrahedron Letters* 41(10):1531-1533; and Gilon et al., 1967, *Tetrahedron* 23(11):4441-4447. The aminoxy group reacts with a reactive group on the Ligand unit.



wherein —R⁹ is selected from the group consisting of —C₁-C₁₀ alkylene-, —C₃-C₈ carbocyclo-, -arylene-, —C₁-C₃₀ heteroalkylene-, —C₃-C₈ heterocyclo-, —C₁-C₁₀ alkylene-arylene-, -arylene-C₁-C₁₀ alkylene-, —C₁-C₁₀ alkylene-(C₃-C₈ carbocyclo)-, —(C₃-C₈ carbocyclo)-C₁-C₁₀ alkylene-, —C₁-C₁₀ alkylene-(C₃-C₈ heterocyclo)-, and —(C₃-C₈ heterocyclo)-C₁-C₁₀ alkylene-.

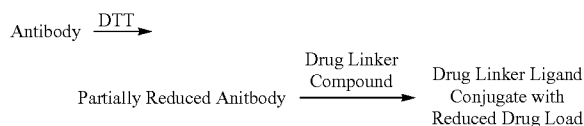
Isothiocyanate Stretchers of the formula shown below may be prepared from isothiocyanatocarboxylic acid chlorides as described in *Angew. Chem.*, 87(14):517 (1975). The isothiocyanate group reacts with a reactive group on the Ligand unit.



wherein —R⁹— is as described herein.

Conjugation of Drug Linker Compounds to Ligand Units
Scheme 4 illustrates methodology useful for making Drug Linker Ligand conjugates having about 2 to about 4 drugs per Ligand unit, as exemplified by an antibody. An antibody is treated with a reducing agent, such as dithiothreitol (DTT) to reduce some or all of the interchain cysteine disulfide residues to form highly nucleophilic cysteine thiol groups (—CH₂SH). The partially reduced antibody thus reacts with Drug Linker compounds, or Linker unit reagents, with electrophilic functional groups such as maleimide or α-halo carbonyl, according to the conjugation method at page 766 of Klussman et al., 2004, *Bioconjugate Chemistry* 15(4): 765-773.

Scheme 4



For example, an antibody, dissolved in 500 mM sodium borate and 500 mM sodium chloride at pH 8.0, is treated with an excess of 100 mM dithiothreitol (DTT). After incubation at 37° C. for about 30 minutes, the buffer is

In Vivo Efficacy of Selected
Auristatin-Dipeptide-hBU12 Conjugates in a
Lymphoma Xenograft Model

The efficacies of Auristatin-Dipeptide hBU12 conjugates with the aryl and alkyl-maleimide Stretcher units were evaluated in a DoHH-2 follicular B-cell lymphoma subcutaneous model. Auristatin-Dipeptide-hBU12 conjugates with average of 4 drug moieties per antibody were used. DoHH-2 cells were implanted subcutaneously into immunodeficient SCID mice (5×10^6 cells per mouse). Tumor volumes are calculated using the formula $(0.5 \times L \times W^2)$ where L and W are the longer and shorter of two bidirectional measurements. Treatment was initiated when average size of tumors reached 100 mm^3 ; the treatment schedule was q4d $\times$ 4 at 3 mg/kg. The results of this study are shown in FIG. 16.

Conjugates of aryl and alkyl maleimide analogues of corresponding Auristatin-Dipeptides showed similar efficacies. AM-Asn-(D)Lys ADCs were more potent than corresponding AW-Met-(D)Lys conjugates.

Example 74

In Vivo Efficacy of Auristatin-Dipeptide-cAC10
Conjugates in a Lymphoma Xenograft Model

The efficacy of selected Auristatin-Dipeptide-cAC10 conjugates having Stretcher units with aryl or alkyl maleimides were evaluated in L540cy Hodgkin's lymphoma subcutaneous model. Auristatin-Dipeptide-cAC10 conjugates with average of 4 drug moieties per antibody were used. L540cy cells were implanted subcutaneously into immunodeficient SCID mice (5×10^6 cells per mouse). Tumor volumes were calculated using the formula $(0.5 \times L \times W^2)$ where L and W are the longer and shorter of two bidirectional measurements. Treatment was initiated when average size of tumors reached 100 mm^3 . The treatment schedule was q4d $\times$ 3 at 1 mg/kg. The results of this study are shown in FIG. 17. The conjugates showed similar efficacy. AM-Asn-(D)Lys ADCs were more potent than corresponding AW-Met-(D)Lys conjugates in this study.

Example 75

Plasma Clearance in Rat

Plasma clearance pharmacokinetics of antibody drug conjugates and total antibody is studied in Sprague-Dawley rats (e.g., from Charles River Laboratories, 250-275 grams each). Animals are dosed by bolus tail vein injection (IV Push). Approximately 300 μl whole blood is collected through jugular cannula, or by tail stick, into lithium/heparin anticoagulant vessels at each timepoint: 0 (predose), 10, and 30 minutes; 1, 2, 4, 8, 24 and 36 hours; and 2, 3, 4, 7, 14, 21, and 28 days post dose. Total antibody is measured by ELISA, for example, by coating with the extracellular domain of the target protein and detecting with an anti-human Fc-HRP antibody conjugate (ECD/G $\times$ huFc-HRP). Antibody drug conjugate is measured by ELISA, for example, by coating with an anti-drug or antiFc antibody and detecting with an extracellular domain-biotin conjugate and a streptavidin-horse radish peroxidase conjugate.

Plasma Clearance in Monkey

Plasma clearance pharmacokinetics of antibody drug conjugates and total antibody can be studied in cynomolgus monkeys, using a similar procedure to that described above in Example 75.

All references cited herein are incorporated by reference in their entirety and for all purposes to the same extent as if each individual publication or patent or patent application was specifically and individually indicated to be incorporated by reference in its entirety for all purposes. The recitation of any reference in this application is not an admission that the reference is prior art to this application.

What is claimed is:

1. A compound having the formula:



or a pharmaceutically acceptable salt or solvate thereof; wherein L is Ligand Unit, LU is a Linker unit, and D is a Drug unit;

L is a peptide, polypeptide or protein that specifically binds to an antigen;

LU has the formula $—W_w-A_a-$,

W_w is a sequence of w independently selected amino acid diradicals, wherein the W proximal to the Drug unit (W_1) is a non-natural amino acid linked via a peptide bond to the Drug unit and the remaining W_s are D-amino acids, provided that W_1 cannot form secondary amide with C-terminal amino acid of D and that the peptide bond can be cleaved by an intracellular protease,

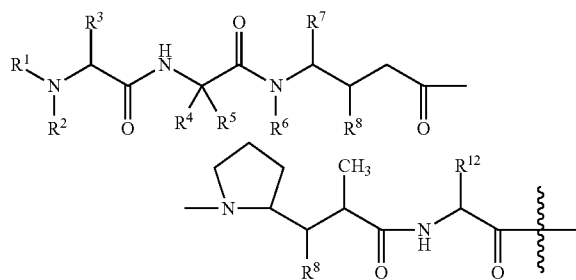
w is an integer ranging from 1 to 12,

A is Stretcher unit, and

a is 1 or 2;

p is an integer ranging from 1 to 20; and

D has the formula:



wherein the wavy line indicates the peptide bond to LU; R^1 and R^2 are independently selected from the group consisting of $—H$ and $—C_1-C_8$ alkyl, with the proviso that both R^1 and R^2 are not $—H$;

R^3 is selected from the group consisting of $—H$, $—C_1-C_8$ alkyl, $—C_3-C_8$ carbocycle, -aryl, $—C_1-C_8$ alkyl-aryl, $—X^1-(C_3-C_8$ carbocycle), $—C_3-C_8$ heterocycle and $—X^1-(C_3-C_8$ heterocycle);

R^4 is selected from the group consisting of $—H$, $—C_1-C_8$ alkyl, $—C_3-C_8$ carbocycle, -aryl, $—X^1$ -aryl, $—C_1-C_8$ alkyl- $(C_3-C_8$ carbocycle), $—C_3-C_8$ heterocycle and $—X^1-(C_3-C_8$ heterocycle);

R^5 is selected from the group consisting of $—H$ and methyl,

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or R^4 and R^5 jointly form a carbocyclic ring and have the formula $-(CR^aR^b)_n-$ wherein R^a and R^b are independently selected from the group consisting of $-H$ and $-C_1-C_8$ alkyl and n is selected from the group consisting of 2, 3, 4, 5 and 6;

R^6 is selected from the group consisting of $-H$ and $-C_1-C_8$ alkyl;

R^7 is selected from the group consisting of $-H$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle, $-aryl$, $-X^1-aryl$, $-X^1-(C_3-C_8$ carbocycle), $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle);

each R^8 is independently selected from the group consisting of $-H$, $-OH$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle and $-O-(C_1-C_8$ alkyl);

R^{12} is the side chain of a natural (L) amino acid selected from the group consisting of arginine, glutamine, tyrosine, lysine, methionine, glycine, alanine, histidine, serine, glutamic acid, aspartic acid, threonine, leucine, asparagine, isoleucine, tryptophan, phenylalanine and valine; and

each X^1 is independently $-C_1-C_{10}$ alkylene-.

2. A compound having the formula:

D-LU

or a pharmaceutically acceptable salt or solvate thereof; wherein LU- is a Linker unit and D is a Drug unit;

LU has the formula W_w-A_a- ,

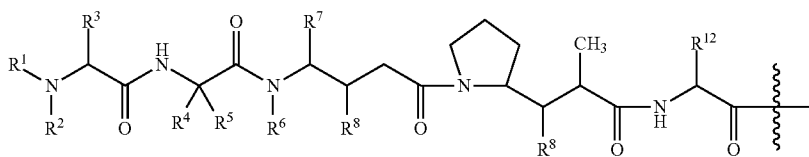
W_w is a sequence of w independently selected amino acid diradicals, wherein the W proximal to the Drug unit (W_1) is a non-natural amino acid linked via a peptide bond to the Drug unit and the remaining W s are D-amino acids, provided that W_1 cannot form secondary amide with C-terminal amino acid of D and that the peptide bond can be cleaved by an intracellular protease,

w is an integer ranging from 1 to 12,

A is Stretcher unit, and

a is 1 or 2; and

D has the formula:



wherein the wavy line indicates the peptide bond to LU; R^1 and R^2 are independently selected from the group consisting of $-H$ and $-C_1-C_8$ alkyl, with the proviso that both R^1 and R^2 are not $-H$;

R^3 is selected from the group consisting of $-H$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle, $-aryl$, $-C_1-C_8$ alkyl-aryl, $-X^1-(C_3-C_8$ carbocycle), $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle);

R^4 is selected from the group consisting of $-H$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle, $-aryl$, $-X^1-aryl$, $-C_1-C_8$ alkyl- $(C_3-C_8$ carbocycle), $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle);

R^5 is selected from the group consisting of $-H$ and methyl,

or R^4 and R^5 jointly form a carbocyclic ring and have the formula $-(CR^aR^b)_n-$ wherein R^a and R^b are independently selected from the group consisting of $-H$ and $-C_1-C_8$ alkyl and n is selected from the group consisting of 2, 3, 4, 5 and 6;

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R^6 is selected from the group consisting of $-H$ and $-C_1-C_8$ alkyl;

R^7 is selected from the group consisting of $-H$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle, $-aryl$, $-X^1-aryl$, $-X^1-(C_3-C_8$ carbocycle), $-C_3-C_8$ heterocycle and $-X^1-(C_3-C_8$ heterocycle);

each R^8 is independently selected from the group consisting of $-H$, $-OH$, $-C_1-C_8$ alkyl, $-C_3-C_8$ carbocycle and $-O-(C_1-C_8$ alkyl);

R^{12} is the side chain of a natural (L) amino acid selected from the group consisting of arginine, glutamine, tyrosine, lysine, methionine, glycine, alanine, histidine, serine, glutamic acid, aspartic acid, threonine, leucine, asparagine, isoleucine, tryptophan, phenylalanine and valine; and

each X^1 is independently $-C_1-C_{10}$ alkylene-.

3. The compound of claim 1 or 2, wherein R^{12} is the side chain of tryptophan or methionine.

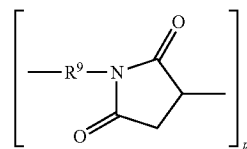
4. The compound of claim 3, wherein R^1 and R^2 are $-CH_3$.

5. The compound of claim 1 or 2, wherein w is an integer ranging from 2 to 12 and the remaining W 's are D-isomers of natural amino acids.

6. The compound of claim 1 or 2, wherein a is 1.

7. The compound of claim 1 or 2, wherein a is 2.

8. The compound of claim 6, wherein A has the formula $-NH-R^9-R^{11}$ wherein R^9-R^{11} have the formula:



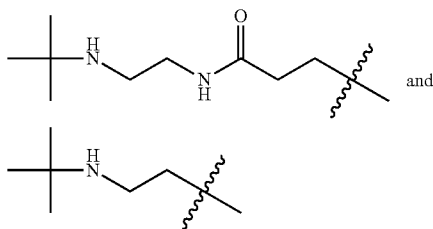
wherein R^9 is selected from the group consisting of $-C_1-C_{10}$ alkylene-, $-C_3-C_8$ carbocyclo-, $-arylene$ -,

$-C_1-C_{30}$ heteroalkylene-, $-C_3-C_8$ heterocyclo-, $-C_1-C_{10}$ alkylene-arylene-, $-arylene-C_1-C_{10}$ alkylene-, $-C_1-C_{10}$ alkylene- $(C_3-C_8$ carbocyclo)-, $-(C_3-C_8$ carbocyclo)- C_1-C_{10} alkylene-, $-C_1-C_{10}$ alkylene- $(C_3-C_8$ heterocyclo)-, and $-(C_3-C_8$ heterocyclo)- C_1-C_{10} alkylene-.

9. The compound of claim 6, wherein $-NH-R^9$ is selected from the group consisting of $-NH-C_1-C_{10}$ alkylene-, $-NH-C_1-C_{10}$ alkylene- $NH-C(O)-C_1-C_{10}$ alkylene-, $-NH-C_1-C_{10}$ alkylene- $C(O)-NH-C_1-C_{10}$ alkylene-, $-NH-(CH_2CH_2O)_r-$, $-NH-(CH_2CH_2O)_r-CH_2-$, $-NH-C_1-C_{10}$ alkylene-, $-NH-C_1-C_{10}$ alkylene- $NH-C(O)-C_1-C_{10}$ alkylene-, $-NH-(C_3-C_8$ carbocyclo)-, $-NH-(arylene)-$, $-NH-C_1-C_{10}$ alkylene- $NH-C(O)-(arylene)-$, and $-NH-(C_3-C_8$ heterocyclo)-, wherein r is an integer ranging from 1 to 10.

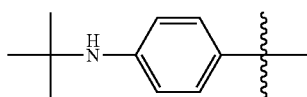
10. The compound of claim 9, wherein $-NH-R^9$ is selected from the group consisting of:

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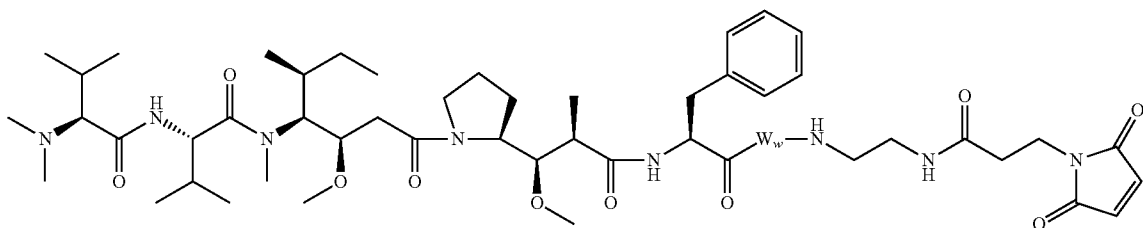
wherein the bar line indicates the point of attachment to a W of W_w and the wavy line indicates the point of attachment to the remainder of A.

11. The compound of claim 9, wherein —NH—R^9 is:



wherein the bar line indicates the point of covalent attachment to a W of W_w and the wavy line indicates covalent attachment to the remainder of A.

12. The compound of claim 10, having the formula:



13. The compound of claim 1 or 2, wherein W_1 is selected from the group consisting of ornithine, penicillamine, β -alanine, aminoalkanoic acid, aminoalkynoic acid, aminoalkanedioic acid, aminobenzoic acid, amino-heterocycloalkanoic acid, heterocyclo-carboxylic acid, citrulline, statine, and diaminoalkanoic acid.

14. The compound of claim 1 or 2, wherein the remaining W_s are independently selected from the D-isomers of the group consisting of alanine, arginine, aspartic acid, asparagine, histidine, glycine, glutamic acid, glutamine, phenylalanine, lysine, leucine, serine, tyrosine, threonine, isoleucine, tryptophan and valine.

15. The compound of claim 14, wherein W_1 is β -Alanine.

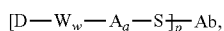
16. The compound of claim 1 or 2, wherein w is 2.

17. The compound of claim 1 or 2, wherein p ranges from 2 to 8.

18. The compound of claim 17, wherein p ranges from 2 to 5.

19. The compound of claim 1, wherein L is an antibody.

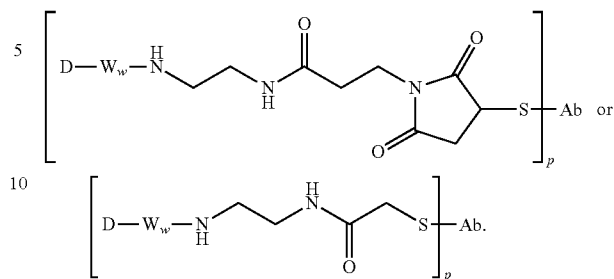
20. The compound of claim 19, wherein the antibody (Ab) is attached to each Amino Acid unit W_w through a cysteine residue of the antibody and the compound has the following formula:



wherein S is a sulfur atom of a cysteine residue of the antibody.

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21. The compound of claim 20, having the formula:



22. The compound of claim 19, wherein the antibody is a monoclonal antibody.

23. The compound of claim 19, wherein the antibody is a bispecific antibody.

24. The compound of claim 19, wherein the antibody is a humanized antibody.

25. The compound of claim 19, wherein the antibody is an antigen-binding antibody fragment.

26. The compound of claim 19, wherein the antibody specifically binds to a B cell antigen.

27. The compound of claim 1 or 2, or a pharmaceutically acceptable salt or solvate thereof, that is in isolated and purified form.

28. A pharmaceutical composition comprising an effective amount of the compound of claim 1, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable diluent, carrier or excipient.

29. A method for killing or inhibiting the proliferation of tumor cells or cancer cells comprising contacting tumor cells or cancer cells in a patient with an amount of the compound of claim 1, or a pharmaceutically acceptable salt or solvate thereof, said amount being effective to kill or inhibit the proliferation of the tumor cells or cancer cells.

30. A method for treating cancer comprising administering to a patient in need thereof an amount of the compound of claim 1, or a pharmaceutically acceptable salt or solvate thereof, said amount being effective to treat the cancer.

31. The method of claim 30, further comprising administering an effective amount of an additional anticancer agent.

32. A method for killing activated lymphocytes associated with an autoimmune disease comprising administering to a patient with an autoimmune disease an amount of the compound of claim 1, or a pharmaceutically acceptable salt or solvate thereof, the amount being effective to kill activated lymphocytes associated with the autoimmune disease in the patient.

33. A method for treating an infectious viral disease comprising administering to a patient in need thereof an